FRESENIUS KABI

AmiCORE

Apheresis System Service Manual

April 2017 47-19-08-650

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Chapter 1 - Introduction

Section 1.1 General Description

Separator

The AmiCORE Apheresis System is an automated blood cell separator intended for use in the collection of blood components. The AmiCORE Apheresis System utilizes apheresis kits to collect platelets from donors. One apheresis kit is used per collection. Apheresis kits are not reusable.

The AmiCORE Apheresis System is designed to collect products while maintaining an extracorporeal volume at or below 15% of total blood volume and a donor post-platelet count greater than or equal to 100,000 platelets/uL.

The AmiCORE Apheresis System is Product Code 6R8800.

Apheresis Kits

AmiCORE apheresis kits are the only apheresis kits that are to be used with the AmiCORE Apheresis System.

The AmiCORE Apheresis System may collect platelets pheresis, leukocytes reduced (single or double units).

In countries where it has been cleared by the appropriate regulatory agencies: platelets pheresis leukocytes reduced may be stored in the platelet storage container(s) for up to five days at 20 degrees Celsius to 24 degrees Celsius with continuous gentle agitation.

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Chapter 2 - Principles of Operation

Section 2.1 Introduction

This chapter covers the principles of operation for the AmiCORE Apheresis System. It provides an introduction to the hardware, disposable kit, and service user interface.

This section contains the following topics pertaining to the AmiCORE Apheresis System:

- Instrument Overview
- Disposable Overview
- Procedure Overview
- External System Overview
- Internal System Overview
- Service User Interface Overview
- Interconnection Diagram

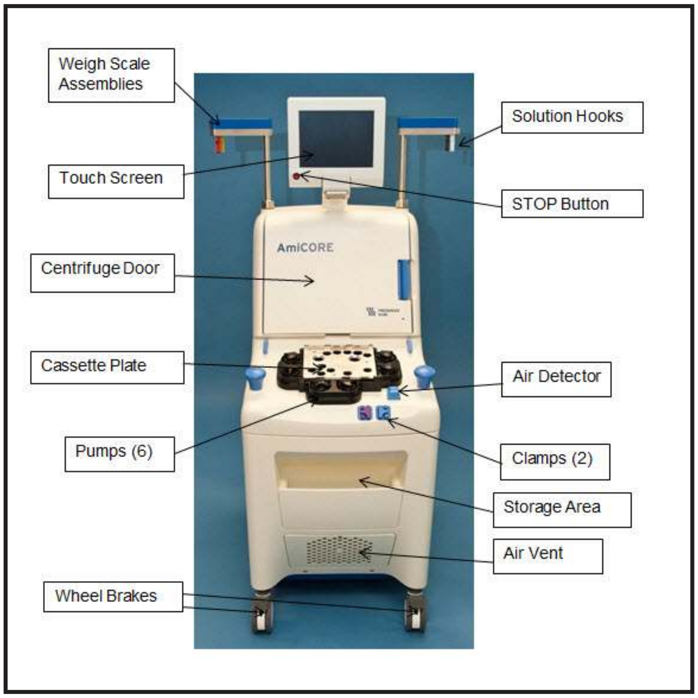
Section 2.2 Instrument Overview

The AmiCORE Apheresis System is a microprocessor-controlled, electromechanical instrument capable of separating donor whole blood and collecting leukoreduced platelets. The collected platelets are stored in a platelet storage container, located within the apheresis disposable set, and stored in a quantity of donor plasma.

The instrument features:

- Touch Screen Display and User Interface
- Two Weight Scales

- Two Solution Hooks
- Six Peristaltic Pumps
- Two Tubing Clamps
- One Air Detector
- Four Pressure Sensors
- Centrifuge Assembly
- Two Solution Poles
- Nine Cassette Module Valves
- Two USB Ports
- Wireless and Wired Communication
- Interface Detector System
- STOP Button
- Leak Detector

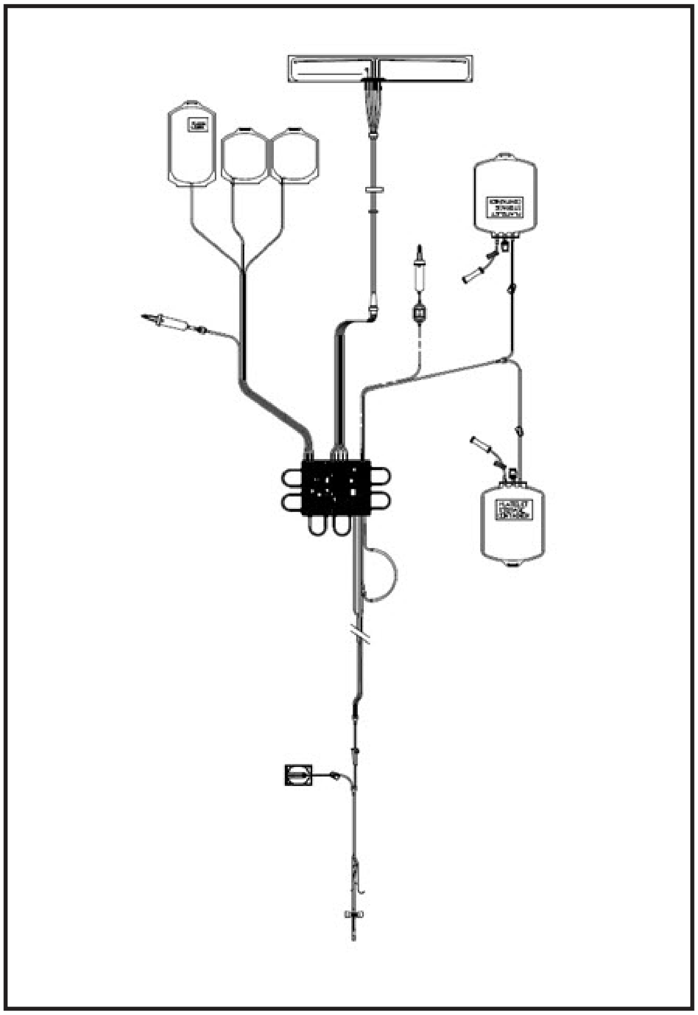

Weigh Scale Assemblies
Solution Hooks
Touch Screen
STOP Button
AmiCORE
Centrifuge Door
Cassette Plate
Air Detector
Pumps (6)
Clamps (2)
Storage Area
Air Vent
Wheel Brakes


Daily Power Switch
Wi-Fi Antenna
Data Ports (Ethernet and USB)
Pressure Cuff Connection
Cassette Bypass Button
Power Cord Connection
Main Power Switch

Section 2.3 Disposable Overview

The AmiCORE Apheresis System utilizes disposable apheresis kits to process donor blood and collect platelets. The disposable apheresis kit contains the following components:

- Two spikes for the attachment of ACD and saline
- One or two platelet collection containers
- Fistula
- Sample Pouch
- Cassette
- Centrifuge Pack and Umbilicus Assembly
- Containers (Plasma, Red Blood Cell, In-Process)



2.1 AmiCORE Apheresis System Disposable Kit (Two Platelet Containers)

Section 2.4 Procedure Overview

The AmiCORE Apheresis System provides a single needle procedure for the collection of donor platelets. The system is configurable based on donor characteristics (weight, height, hematocrit, platelet pre-count) and may be used to collect a single, double, or triple platelet product (based on donor eligibility and regional requirements). Blood is processed throughout multiple draw (blood entering AmiCORE) and return (blood returned to donor) cycles.

The process of collecting donor platelets from donor whole blood is a two phase system of separation. Both phases occur within the centrifuge;

each occupying a portion of the centrifuge pack (separation side and collection side).

Phase 1: Whole Blood Separation

The AmiCORE Apheresis System pumps the donor blood into the cassette. The blood is mixed with ACD (ratio based on input variable setting) to prevent coagulation. Blood is pumped into the in-process (IP) bag while simultaneously being pumped from the IP bag into the centrifuge for separation. In this phase of separation the centrifuge is spinning and the anticoagulated whole blood separates into packed red cells and platelet-rich plasma. Both the packed red cells and the platelet-rich plasma are pumped out of the centrifuge. The packed red cells are sent, temporarily, to the RBC container. These cells will be returned during a return cycle. The platelet-rich plasma is pumped back into the centrifuge, into the collection side of the centrifuge pack. This is where the second phase of separation occurs.

Phase 2: Platelet Collection

The platelet-rich plasma is pumped into the centrifuge, into the collection side of the centrifuge pack. In this phase of separation, the centrifuge is spinning and the platelet-rich plasma separates into platelets, which remain in the centrifuge pack, and platelet-poor plasma which is pumped out of the pack and into the plasma container.

The AmiCORE Apheresis System continuously performs the two separation phases throughout the procedure to collect the target volume of platelets. Procedure length is determined by the time to process a specific amount of donor whole blood, as calculated by the input parameters and the target platelet yield.

The final step of a platelet collection, platelet transfer, occurs after the desired whole blood to be processed has been processed and the donor has been disconnected. Platelet-poor plasma is pumped into the collection chamber where the collected platelets are re-suspended by the operator and then transferred to the final collection container(s).

Section 2.5 External System Overview

User Interface

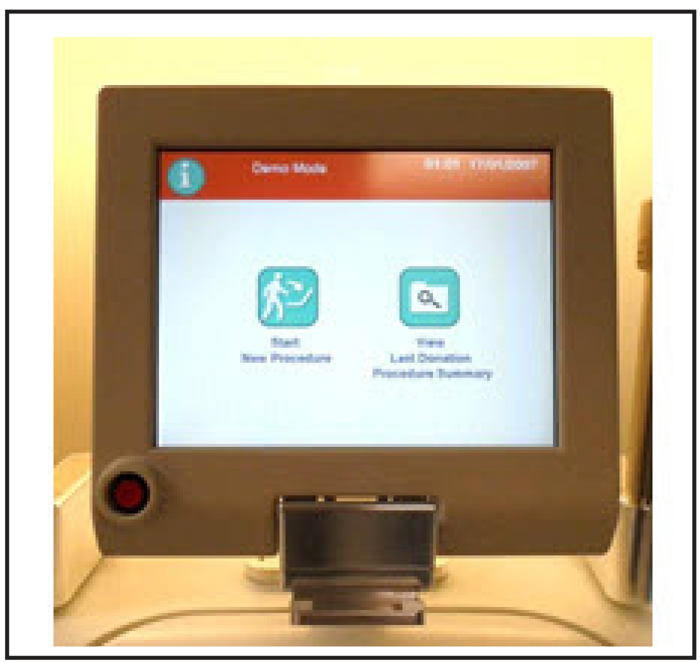
The user interface consists of five areas:

- Touch Screen Display
- STOP Button
- Main Power Switch
- Daily Power Switch
- Speaker

Touch Screen Display

The Touch Screen display is a color display. It is the primary communication interface between the AmiCORE Apheresis System and the operator.

The display is a Touch Screen. When the user touches a specific area of the display screen corresponding to a button, tab, or variable entry, the Touch Screen detects these finger touches and reports the information to the single board computer (SBC) board.



STOP Button

The STOP *button* is located on the front of the instrument just below and to the left of the Touch Screen display. If the instrument is running, pressing the STOP *button* halts the procedure and stops the movement of all pumps and clamps in a controlled manner. The STOP *button* is also utilized to properly power down the instrument.

Main Power Switch

The main power switch is located in the lower rear of the instrument and is the master power switch used to provide power to the instrument. It must be set to ON to operate the instrument. It is used in conjunction with the daily power switch to control power to the AmiCORE Apheresis System.

Daily Power Switch

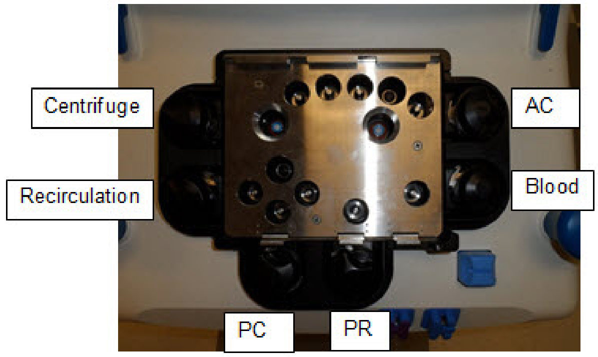
The daily power switch is located in the rear of the touch screen display housing. It must be set to ON to operate the instrument. It is used in conjunction with the main power switch to control power to the AmiCORE Apheresis System. The daily power switch is easier to access than the main power switch and can be used for powering down the instrument each day (following proper shut down procedure).

Speaker

The AmiCORE Apheresis System contains a speaker which is utilized for audio tones during alerts and for additional audible feedback.

Pumps

The AmiCORE Apheresis System contains six pumps. The pumps are located on the front of the instrument on the top panel. The pumps are named: centrifuge, recirculation, PC, PRP, blood, and AC.



2.2 AmiCORE Top Panel Assembly

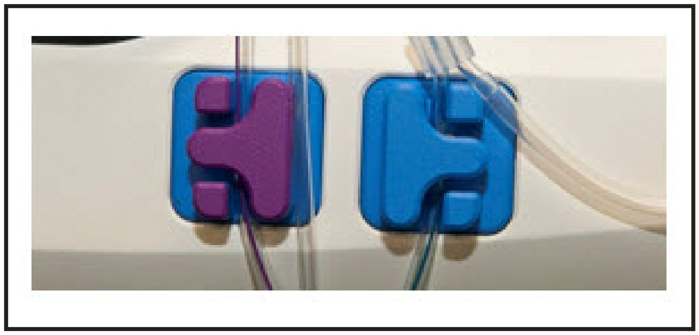
The pumps are responsible for the movement of all fluids within the disposable kit. Each pump provides feedback to the control circuitry.

There are two different gear ratio's utilized for the motors used in the pump assemblies within the AmiCORE Apheresis System. The blood pump, centrifuge pump, and PC pump are 24:1 gear ratio. The AC pump, recirculation pump, and the PRP pump are 71:1 gear ratio. Other than the motor gear ratio, each pump assembly is exactly the same in construction, control, and connection. All pump heads are identical.

The speed of each pump is measured and controlled using tachometer counts from the motor and is redundantly monitored using head sensor counts. These values are converted by the AmiCORE Apheresis System software into pump flow rates.

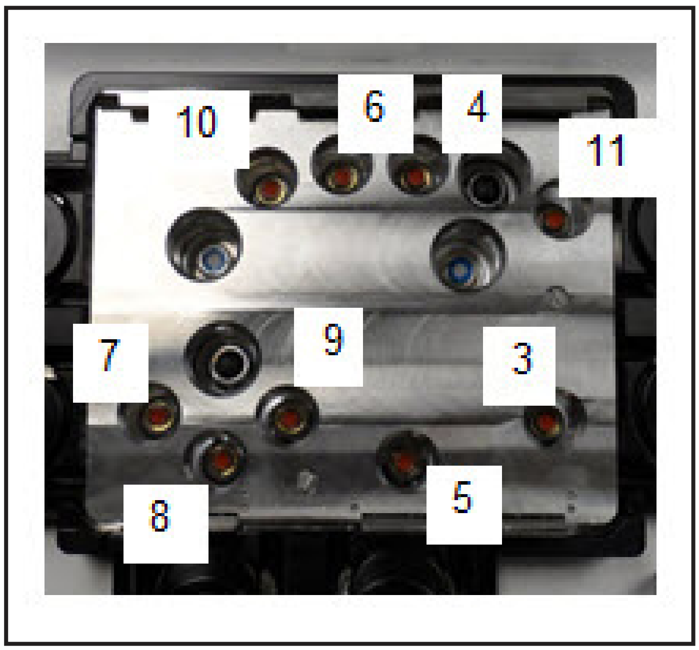
Clamps

The AmiCORE Apheresis System contains two donor clamps (#1 – purple and #2 – blue). The clamps are color blue and located along the front right edge of the top panel. The clamps are controlled via normally closed solenoids. They are used during the procedure to isolate the tubing going to and from the donor during an alert, stop, or other appropriate times.



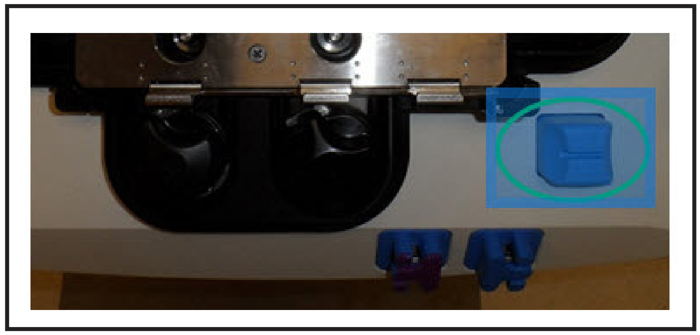
Valves

The AmiCORE Apheresis System contains nine valves. The valves are located within the top panel assembly. The valves are controlled using normally closed solenoids. The purpose of the valves is to occlude pathways within the disposable kit cassette during the procedure in order to direct fluid flow. The valves are numbered for service reference.



Air Detector

The AmiCORE Apheresis System contains an air detector module. The air detector is located in the front right corner of the top panel. The purpose of the air detector is to monitor the disposable tubing line placed in it for air. The disposable tubing line placed in the air detector during the procedure is the donor return line. The air detector monitors this line during the return of fluids. If the air detector detects air during fluid return, it places the AmiCORE Apheresis System in a safe state, closes the donor clamps, raises an alert, and automatically performs an air purge using saline.

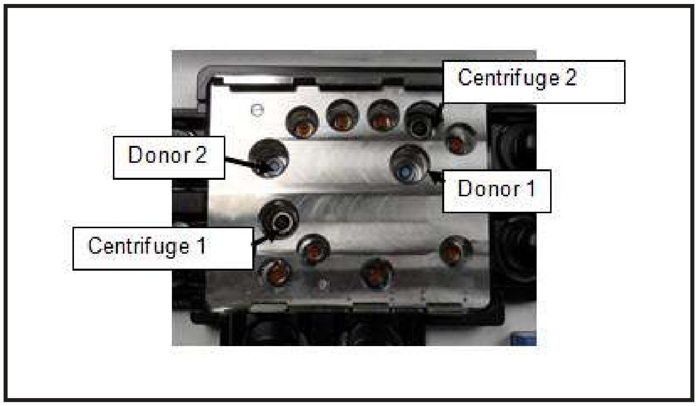


Weight Scales

The AmiCORE Apheresis System contains two weight scale assemblies. The weight scale assemblies are located on the solution pole to the left of the AmiCORE display (when facing the instrument). The weight scales are used to monitor specific fluid levels to and from the instrument during a procedure. Additionally, the AmiCORE contains two solution hooks, located to the right of the AmiCORE display, for hanging saline and final platelet container(s).

Pressure Sensors

The AmiCORE Apheresis System contains four pressure sensors. The pressure sensors are located on the top panel. The purpose of the pressure sensors is to monitor pressures within specific areas of the disposable cassette. There are two donor and two centrifuge pressure sensors. The donor pressure sensors monitor positive and negative pressures. Their primary use is the monitoring of donor vein pressure during the procedure. Additionally donor 2 pressure sensor monitors IP bag pressure. The centrifuge pressure sensors measure positive pressure and monitor the pressure in the separation and collection chambers of the centrifuge pack during the procedure. Their primary use is to monitor centrifuge inlet line occlusions during the procedure.



Network Communications

The AmiCORE Apheresis System contains networking communications hardware which is built into the display/SBC Interface board. This hardware is equipped with two USB ports, an Ethernet port, and a Wi-Fi wireless module. The communications hardware provides the option of using either wireless or wired communications when integrating the AmiCORE Apheresis System with Fresenius Kabi's data management solution.

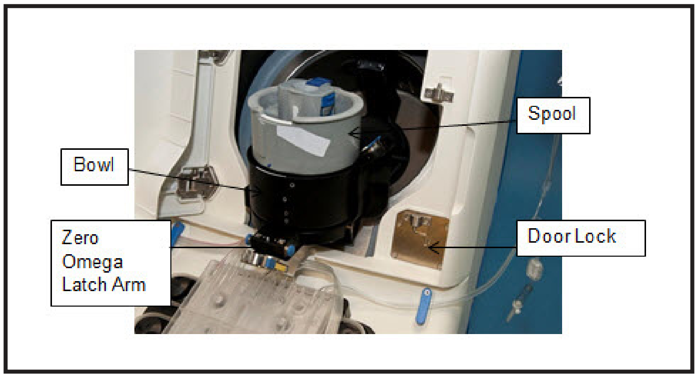
The USB ports may be utilized for the upgrade of AmiCORE Apheresis System software, downloading system setting information, and uploading settings, as well as the export of procedure records and data logs.

Pressure Cuff

The AmiCORE Apheresis System contains a pressure cuff which is attached in the rear of the device. The pressure cuff is periodically inflated and deflated throughout the procedure to help maintain flow rates during draw cycles. The operator can set cuff pressure, in mmHg, using the Touch Screen.

Centrifuge

The AmiCORE Apheresis System contains a centrifuge assembly. The centrifuge is located in the upper portion of the front of the device. The centrifuge door may be opened (when not in a locked state) by pulling on the blue door handle on the right side of the door. The centrifuge compartment contains the centrifuge assembly, a zero omega latch arm, a leak detector, temperature sensors (embedded on the internal wall of compartment), and the interface detector window. Additionally, the centrifuge assembly contains a spool and spool holder.



Section 2.6 Internal System Overview

This section provides an overview of the internal components of the AmiCORE Apheresis System and an overview of the interconnection of subsystems and components.

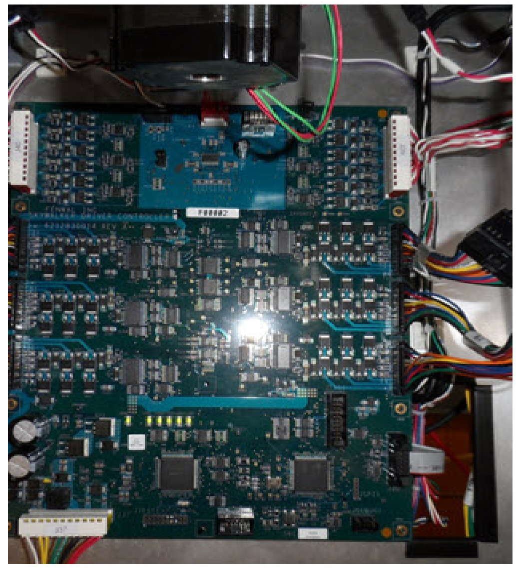
The major printed circuit board assemblies (PCBA) making up the system are:

- Display/Single Board Computer (SBC) Interface PCBA
- Driver Controller PCBA
- Main Controller PCBA
- Interface Controller PCBA
- Centrifuge Driver PCBA
- Weight Scale PCBA
- Power Supply Distribution and Battery Charger PCBA
- Leak Detector and Temperature PCBA
- P1 and P2 Interface PCBA

Display/Single Board Computer Interface PCBA

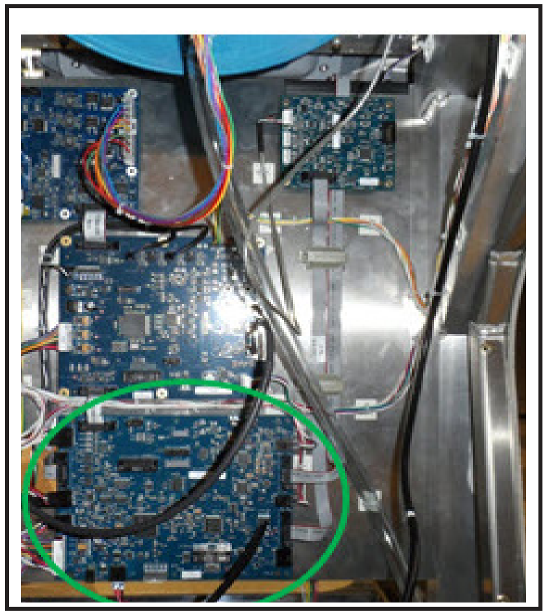
Board	Display/Single Board Computer Display
Part Number	6412830015
Location	Rear above centrifuge bucket
Purpose	Primary computer board containing AmiCORE software. Control board for Touch Screen. Communication hardware (USB ports, Wi-Fi, Ethernet). Primary interface to Main Controller PCBA.
Connections	Display Assembly (J77) Main Controller PCBA (J84) Power (J86) Speaker (J88)

Driver Controller PCBA



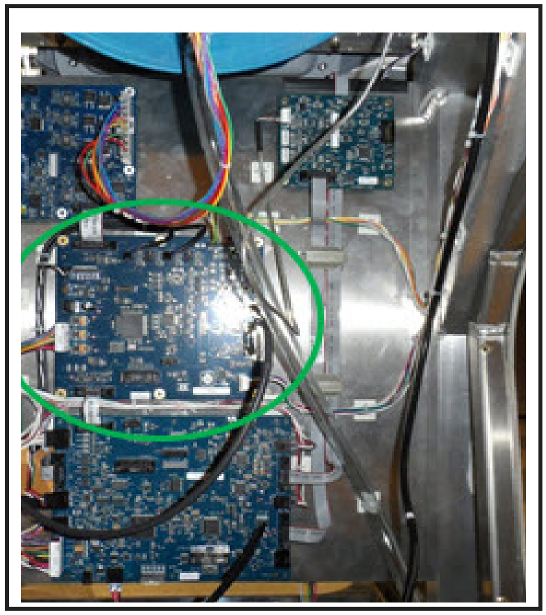
Board	Driver Controller PCBA
Part Number	6412830014
Location	Front below top panel.
Purpose	Primary controller board for most top panel hardware including pumps, clamps, valves, cassette motor.
Connections	Clamp 1-7 (J40) Clamp 8-11 (J39) Cassette Bypass Button (J38) Cassette Motor (J44) Centrifuge Pump (J36) Recirculation Pump (J35) PC Pump (J34) AC Pump (J31) Blood Pump (J32) PRP Pump (J33) Main Controller (J30) Power Supply (J37)

Main Controller PCBA



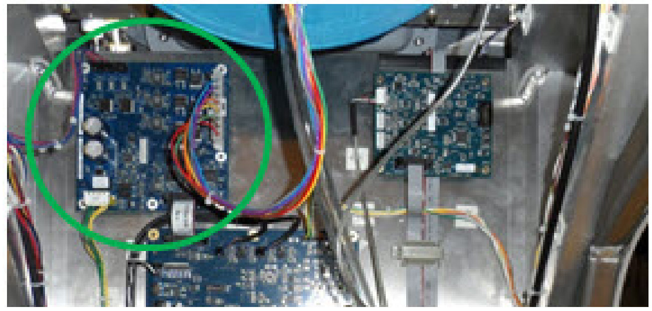
Board	Main Controller
Part Number	6412830004
Location	Rear center divider lowest board.
Purpose	Primary system controller board. Information conduit between SBC and subsystem boards. Safety controller. Air detector communication. Pressure sensor communication. Stop switch communication. Compressor controls.
Connections	Single Board Computer (J1) Driver PCBA (J30) Air Detector (J16) STOP Button (J17) Compressor (J7) Weight Scale PCBA (J45) Pressure Sensor #1 (J14) Pressure Sensor PCBA (J13) Pressure Sensor #4 (J15) Interface Controller PCBA (J53)

Interface Controller PCBA



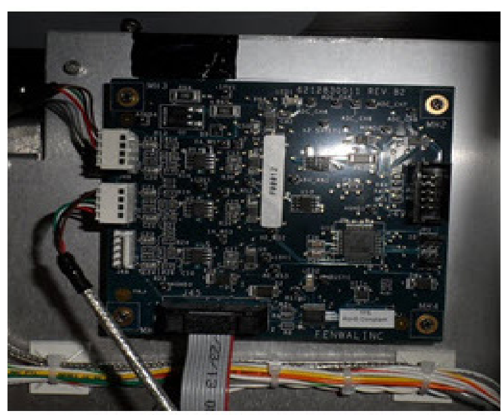
Board	Interface Controller
Part Number	6412830006
Location	Rear center divider middle board.
Purpose	Controller board for interface assembly. Controller board for door sensors and lock mechanism. Communication with leak/temperature board. Communication with centrifuge controller board.
Connections	Centrifuge Controller Board (J56) Door Lock Power (J55) Door Latch Sensor (J57) Door Lock Sensor (J50) Leak/Temperature Board (J51) Fiber Optic #1 (PD1) Fiber Optic #2 (PD2) Fiber Optic #3 (PD3) Main Controller PCBA (J53) Power (J52)

Centrifuge Driver PCBA



Board	Centrifuge Driver
Part Number	6412830007
Location	Rear center divider top left board.
Purpose	Driver board for centrifuge assembly.
Connections	Fan (J62) Fan (J63) Centrifuge Assembly (J61) Interface Controller PCBA (J60) Power (J65)

Weight Scale PCBA

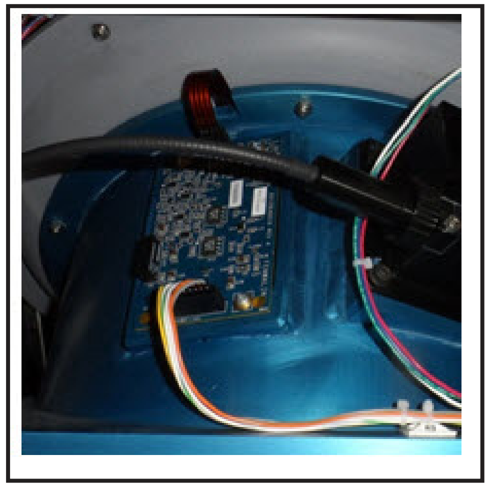


Board	Weight Scale PCBA
Part Number	6412830011
Location	Rear center divider top right board.
Purpose	Interface board for weight scale assemblies.
Connections	Weight Scales (J46 & J47) Main Controller PCBA (J45)

Power Supply Distribution and Battery Charger PCBA

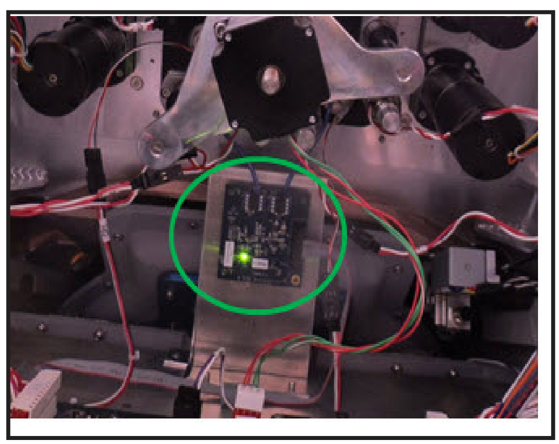
Board	Power Supply Distribution and Battery Charger
Part Number	6412830012
Location	Rear on top of power supply compartment.
Purpose	Power distribution of power supply voltages to system circuit boards.
Connections	Driver Controller PCBA (J92) Centrifuge (J93) Main Controller PCBA (J91) Interface PCBA (J98) Display PCBA (J98) Power Supply (rear) (J21) Display Switch (J97) Power Supply (front) (J96) Power Supply (rear) (J120) Battery terminals (+ & –)

Leak Detector and Temperature PCBA



Board	Leak Detector and Temperature PCBA
Part Number	6412830010
Location	Rear top behind fan frame assembly.
Purpose	Leak detector interface. Temperature sensor interface for centrifuge compartment.
Connections	Interface Detector Interface Controller Board (J82)

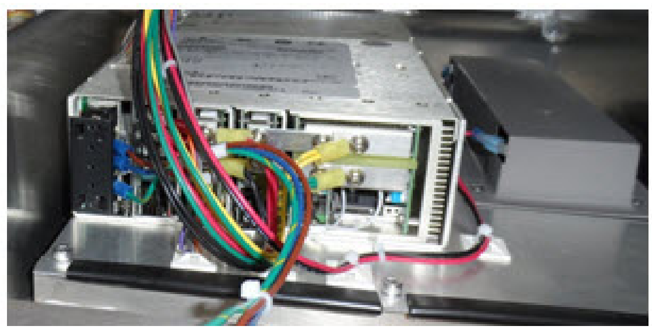
P1 and P2 Interface PCBA



Board	P1 and P2 Interface PCBA
Part Number	6412830008
Location	Front below top panel.
Purpose	Controller board for pressure sensors P1 and P2.
Connections	Pressure Sensor #1 Pressure Sensor #2

Power Supply

The AmiCORE Apheresis System power supply provides the power to the internal circuit boards and components using the power distribution and battery charger PCBA. The AmiCORE power supply provides the following DC voltages: +5V, +15V, –15V, and +24V. The power supply is located in a compartment in the base of the AmiCORE Apheresis System. The power supply compartment additionally contains the AmiCORE Apheresis System backup battery.

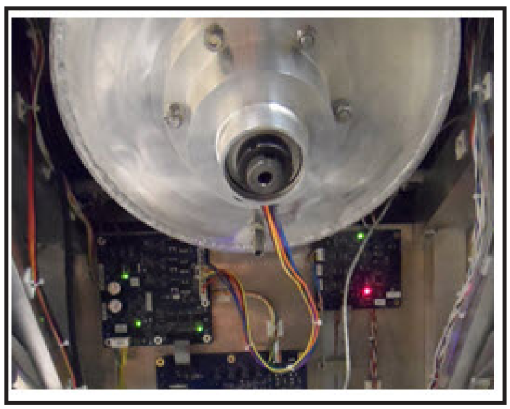


Compressor Assembly

The AmiCORE Apheresis System contains a pneumatic compressor assembly. The compressor assembly is used to inflate and deflate the pressure cuff during AmiCORE procedures. The compressor assembly connects to the main controller PCBA. The compressor is located in the rear of the instrument. The compressor assembly contains a valve for venting the cuff.

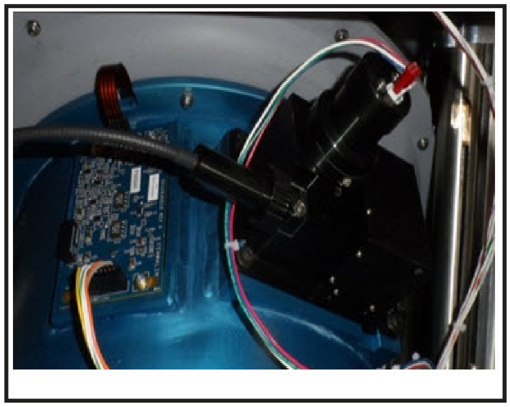
Centrifuge Assembly

The AmiCORE Apheresis System contains a centrifuge assembly. The centrifuge assembly consists of a centrifuge motor, centrifuge, spool, and spool holder.



Interface Detector Assembly and Leak Detector/Temperature PCBA

The top rear compartment of the AmiCORE Apheresis System contains an interface detector assembly and a leak detector/temperature PCBA. The interface detector assembly contains three fiber optic receiving units as well as the interface signal LED. The leak detector/temperature PCBA receives the leak detector signal as well as contains the sensors which monitor centrifuge compartment temperature.



Section 2.7 Service User Interface Overview

This section discusses the user interface for each of the Service Mode menu screens.

Service Mode is a specific set of menus and controls for utilization during the service, installation, and testing of an AmiCORE device. The AmiCORE Apheresis System should only be placed in Service Mode by trained service personnel. Service Mode is password protected. See Section 3.4 for instructions on accessing Service Mode.

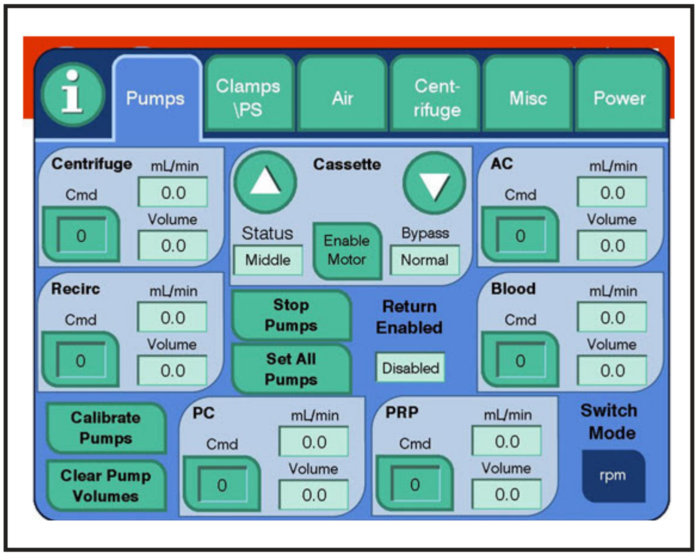
Service Mode consists of six primary tabs. Each tab is accessible by touching the corresponding tab. The following tabs are accessible within Service Mode:

- Pumps
- Clamps\PS
- Air
- Centrifuge
- Misc
- Power

Additional features may be accessed using the *information button* (I-button) in Service Mode.

Note: Hitting the STOP *button* in Service Mode will stop all pumps, close all clamps, stop the centrifuge, and reset all settings to the default.

Pumps Tab



The *pumps tab* contains the following:

- Individual controls for each of the six AmiCORE Apheresis System pumps. Each pump may be set to a commanded speed by selecting the *Cmd field* for the corresponding pump. Speed may be entered by keypad entry. The placement of the pumps on the *pumps tab* correlates to the placement of the actual pump on the AmiCORE top panel.

  Positive speeds correspond to clockwise pump rotation. Negative speeds correspond to counterclockwise pump rotation. In RPM mode, accessible using the *rpm button*, both the motor and pump head speeds in rpm at the pump head will be displayed. In mL/min mode (shown above), the mL/min rate will be displayed along with the tracked volume.

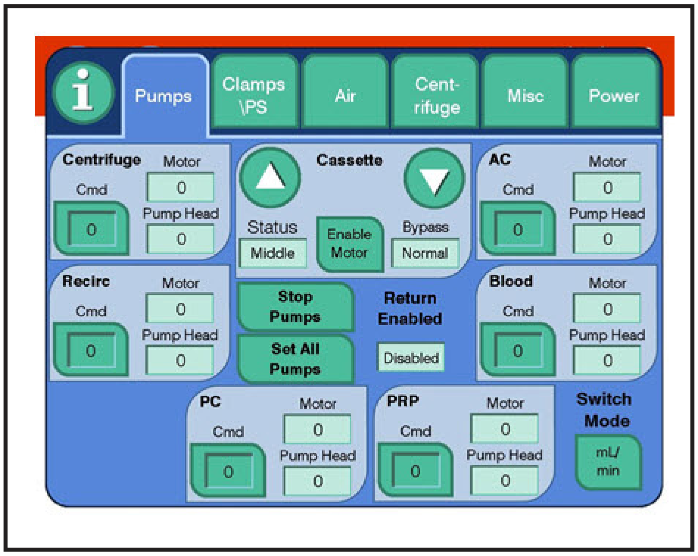
  The allowable range for pump speeds

Centrifuge, PC, AC:	–175 rpm to 175 rpm
	–150 mL/min to 150 mL/min
Recirc, PRP, AC:	–59 rpm to 59 rpm
	–23.5 mL/min to 23.5 mL/min

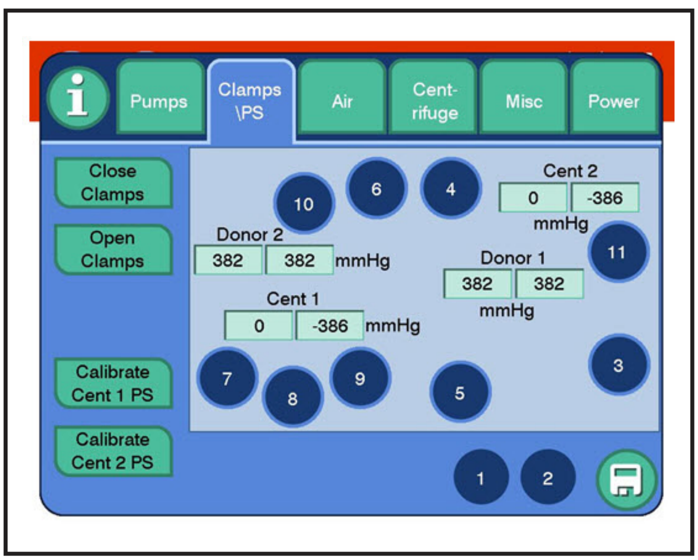
- The cassette plate control can be enabled or disabled using the *enable motor/disable motor button*.

- The cassette plate can be raised and lowered using the *up* and *down arrow*. The *status field* reflects the cassette plate sensor reading: loaded when down, unloaded when up, middle when stopped in-between.

- The *stop pumps button* will stop all pumps.
- The *set all pumps button* will set all pumps to the same commanded speed. The allowable range is –59 rpm to 59 rpm or -23.5 mL/min to 23.5 mL/min.
- The *calibrate pumps*, *clear pump volumes* (only visible in mL/min mode), and *switch mode button* are used to simulate pump calibration in Service Mode.
- *Pumps tab* when RPM mode selected:



Clamps\PS Tab



The *clamps\PS tab* contains the following:

- Individual control for each valve and clamp. The donor clamps are indicated by “1” and “2”. The cassette valves are indicated by “3”, “4”, “5”, “6”, “7”, “8”, “9”, “10”, and “11”. Each numbered button may be selected to either close or open the corresponding valve or clamp. The screen layout corresponds to the valve/clamp location on the top panel. The color of the valve button reflects its state: dark blue is closed, and green is open.

- The *close clamps button* will close all valves/clamps.

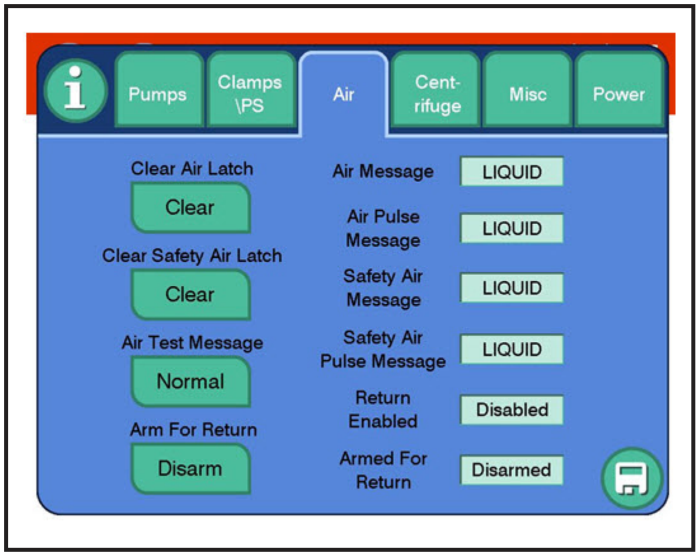
- The *open clamps button* will open all valves/clamps.

- Each pressure sensor output has two values displayed on-screen. These outputs are a live reading of the pressure sensors. The displayed value on the left corresponds to the primary driver reading of the corresponding pressure sensor. The displayed value on the right corresponds to the safety reading of the corresponding pressure sensor.

- The *calibrate cent 1 PS* and *calibrate cent 2 PS* buttons are used in the calibration of the corresponding pressure sensors. See Chapter 6 for additional information on calibration.

- The *computer disc button* in the lower right hand corner of the screen is used to close and zip the current data log file and start a new log file. When the current log file is zipped, a pop-up will notify the operator. This option should be utilized prior to downloading log files of activities performed within Service Mode as the unzipped current log will not be downloaded. This option is found in several service screen tabs. Note that log files are continuously being recorded in Service Mode. Older log files will be overwritten if the instrument is left in Service Mode for extended periods of time.

Air Tab



The *air tab* contains the following:

- The *clear air latch button* is used to clear the latched values of the primary air detector channel readings. The values of the primary air detector channel reading are updated in the *air message* and *air pulse message output fields* (the two fields represent digital and pulse signals respectively).

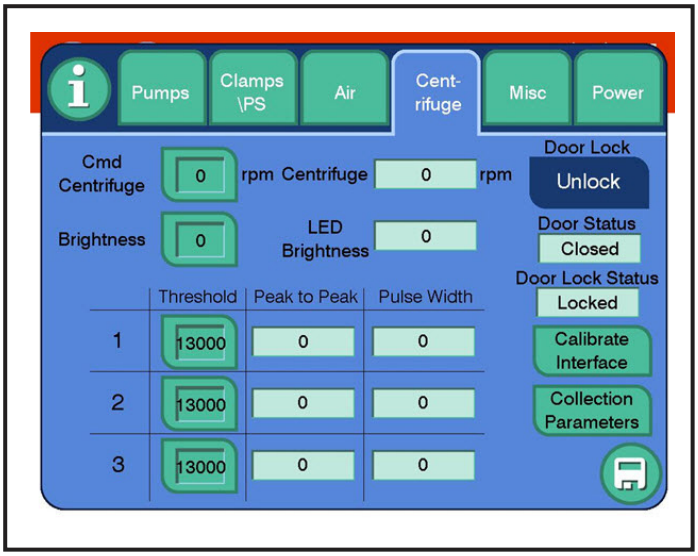
- The *clear safety air latch button* is used to clear the latched values of the safety air detector channel readings. The values of the safety air detector channel reading are updated in the *safety air message* and *safety air pulse message output fields* (the two fields represent digital and pulse signals respectively).

- The *air test message button* is used to force an air reading to the primary and safety channels, regardless of fluid/air inserted into air detector. Selecting the button changes its status from normal to test. When test is displayed, an air signal should be sent to the primary and safety channel if the air detector is functioning properly.

- The *arm for return button* is used to setup a dependency between the blood pump and the air detector. The *armed for return fiel*d displays the current status of the *arm for return* option. The return enabled will enable return when the *arm for return* is armed AND a fluid filled tubing is in the air detector.

  When *arm for return* is set to arm, a fluid filled tubing must be in place in the air detector for the blood pump to be commanded to a speed less than zero (counterclockwise). If the fluid filled tubing is removed when the blood pump is rotating counterclockwise and the return enabled is armed, the blood pump will stop and shortly thereafter an alert will be triggered if the system is operating properly.

Centrifuge Tab



The *centrifuge tab* contains the following:

- The *cmd centrifuge field* is used to enter a commanded centrifuge speed. The actual centrifuge speed is reflected in the *centrifuge field* in rpm units. This option is disabled until the centrifuge door is closed and locked.

- The *brightness field* is used to manually enter a commanded brightness for the interface detector LEDs.

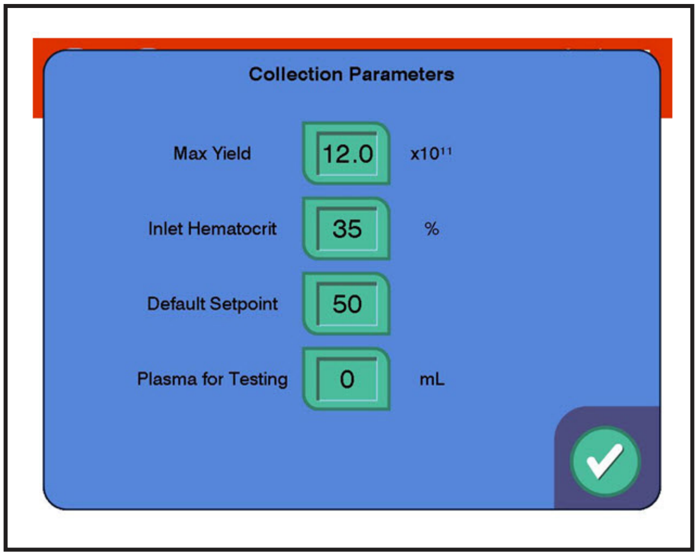
- The *door lock button* is used to lock and unlock the centrifuge compartment door. The centrifuge compartment door lock sensor status is displayed in the *door lock status field*.

- The door sensor indicating door open or closed is displayed in the *door status field*.

- The *calibrate interface button* is used to calibrate the interface signal. See Chapter 6 for additional information on interface calibration.

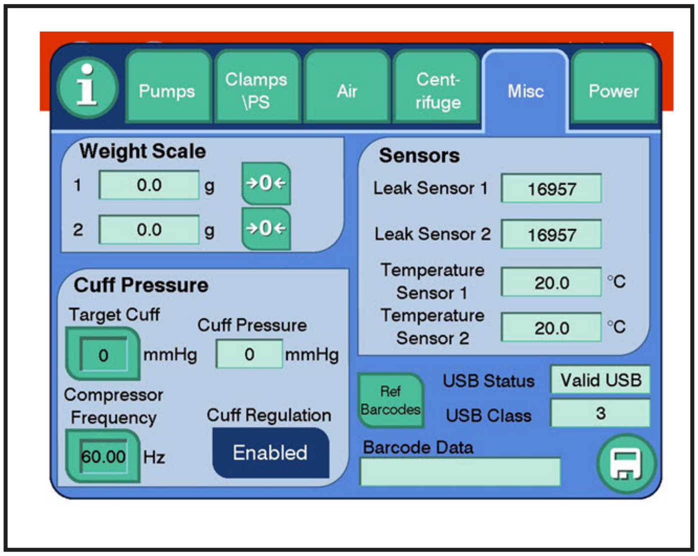
- The *threshold fields* are used for manual entry of interface threshold.

- The *collection parameters button* opens a screen with editable procedure parameters. These values should not be changed without specific instruction.



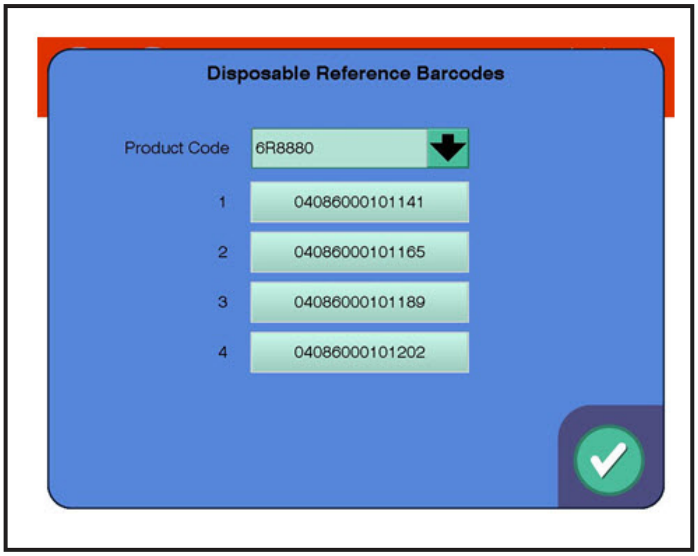
  - The *max yield field* is the maximum targetable yield.
  - The *inlet hematocrit field* is the target hematocrit into the separation chamber.
  - The *default setpoint field* is the separation setpoint.
  - The *plasma for testing field* is used for testing, and therefore is always set to 0 mL.

Misc Tab



The *misc tab* contains the following:

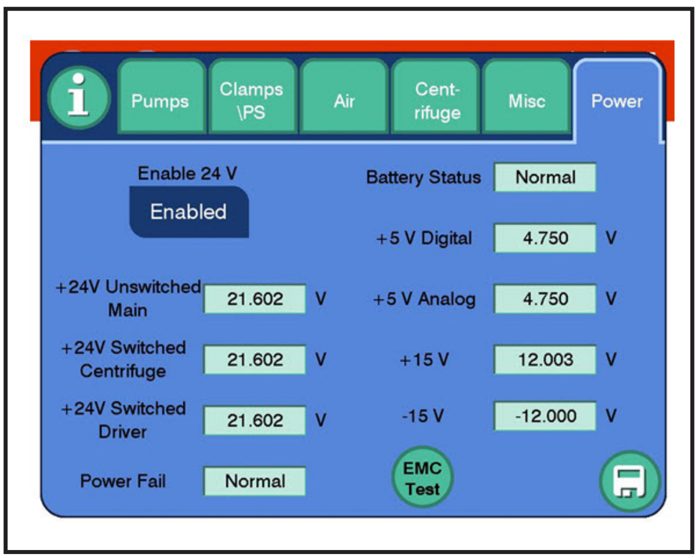
- The weight scale output reading in units grams for weight scale #1 and #2. Scale #1 is the front scale and scale #2 is the rear scale when facing the instrument.
- The option to tare the weight scales using the button with a zero in between two arrows.
- The raw output reading for both leak sensors.
- The output reading for both temperature sensors in units Celsius.
- The *target cuff button* is used to enter a cuff pressure in units mmHG. The actual cuff pressure value is displayed in the *cuff pressure field*.
- The *compressor frequency butto*n is used to manually change the compressor pulse frequency. See Chapter 6 for more details.
- The *cuff regulation button* is used to turn off cuff control. It may be used to lock the current pressure within the cuff to measure the leak rate.
- The *ref barcodes button* will bring up a screen which can be used to enter new kit bar codes for either the 6R8880 kit or the 6R8882 kit. Up to four variations can be stored per kit code. The codes may be entered manually or by scanning a kit bar code.



- The *USB status* and *USB class fields* are used to identify when a USB device is connected to the AmiCORE.

- The *barcode data field* is used to reflect the value of bar codes when scanned with the AmiCORE Apheresis System bar code scanner.

Power Tab



The *power tab* contains the following:

- The *enable 24V button* is used to enable or disable switched 24V.

- The *battery status field* is used to monitor the status of the backup battery. A reading of normal indicates proper battery connection and minimal discharge. A reading of fault indicates that the battery is either disconnected or deeply discharged.

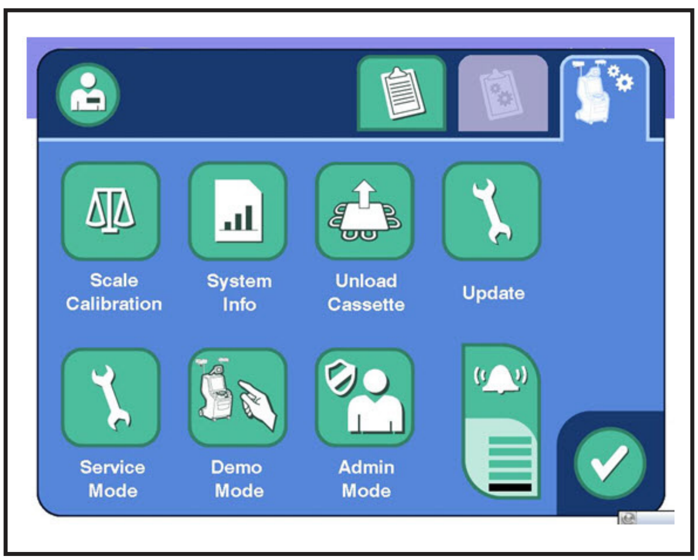
- The *EMC test button* is a development test which displays the readout of several Service Mode sensors. Additionally, it displays the largest delta of readings from the time it is selected.

- The power fail status is used to indicate the status of AC power into the power supply. A reading of normal indicates AC line voltage is present. A reading of fault indicates AC line voltage is absent.

- The power supply voltage values are reflected for the following voltages:

    - +5V Digital
    - +5V Analog
    - +15V
    - -15V
    - +24 Unswitched
    - +24 Switched Centrifuge
    - +24 Switched Driver

I-Button



The *I-button* can access the following:

- Weight Scale Calibration – Calibrate Weight Scales

- Unload Cassette – Manually unload cassette (unavailable in Service Mode).

- Update – Software update using a USB. This option is used to update the AmiCORE software version from a USB drive. Once selected, an overlay will appear and then remove. After this, the device power will need to be cycled two times.

- Service Mode – Access Service Mode using a password.

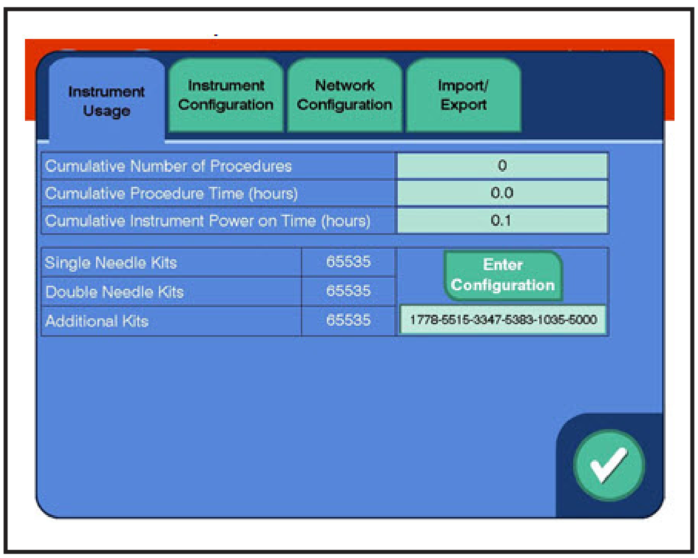
- Admin Mode – Administrative settings. Refer to Administrator's Guide for all Admin Mode settings (unavailable in Service Mode).

- Demo Mode – For non-donor example runs (non-blood procedures – unavailable in Service Mode).

- Alert Volume

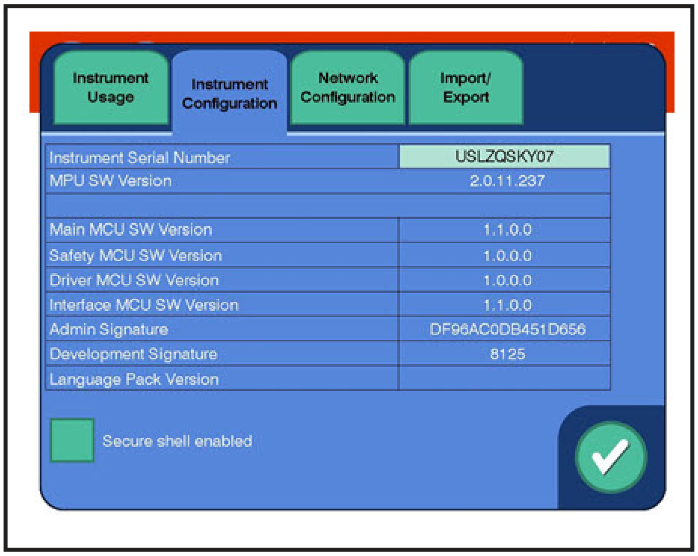
- System Information – Additional screens

Instrument Usage Tab (# Procedures, Procedure Time, Power On Time)



The displayed values are editable in Service Mode.

Instrument Configuration Tab (SW Version, Serial #)

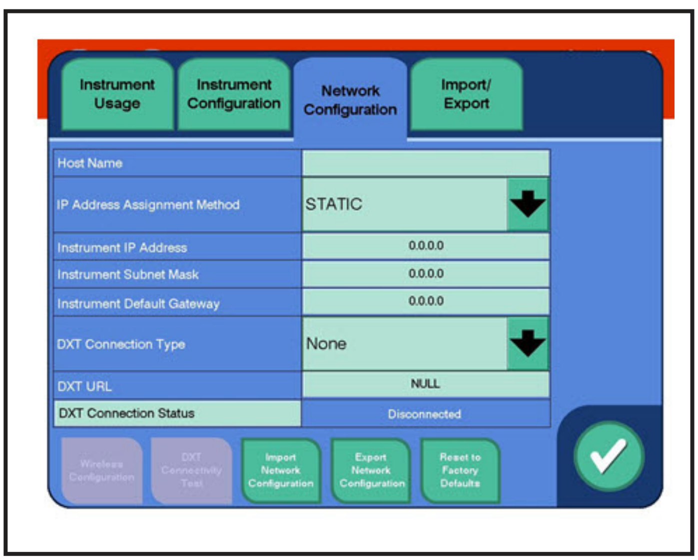


The *instrument serial number field* is editable in Service Mode. If the SBC or flash disk is changed, this should be updated to reflect the instrument serial number.

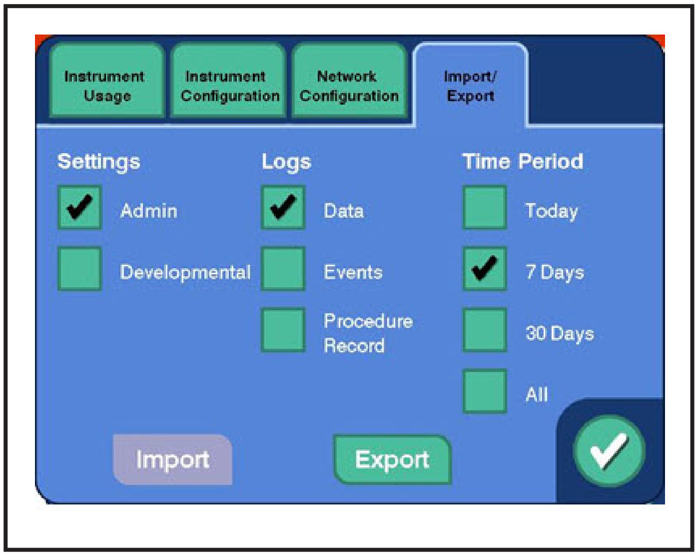
The option to enable/disable secure shell is available. This should be disabled (not checked) for all human use devices outside of controlled clinical trials when instructed.

The full X.X.X.X MPU SW version will be displayed only in Service Mode. Outside of Service Mode, only X.X is displayed.

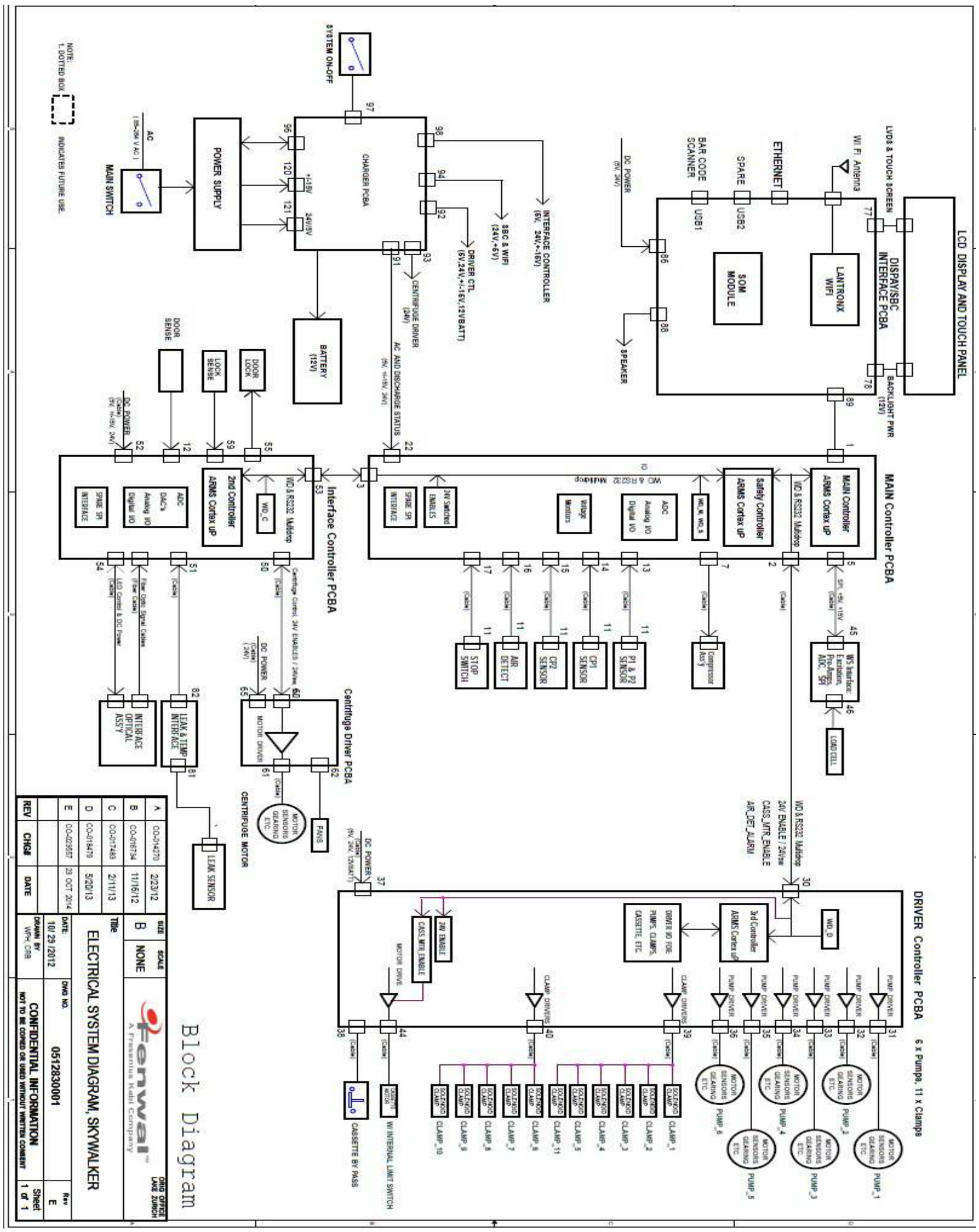
Network Configuration Tab (Wireless/Wired Communication)



Import/Export Tab (Data Downloading from USB)



Section 2.8 Interconnect Diagram



Chapter 3 - Installation and Test Procedures

Section 3.1 Introduction

This section contains the following installation and test procedures for the AmiCORE Apheresis System:

- Unpacking and Assembly
- Visual Inspection
- Power Up System Test
- Enable Service Mode
- Pump Verification
- Cassette Motor Verification
- Clamp Verification
- Valve Verification
- Pressure Sensor Verification
- Air Detector Verification
- Door Lock and Door Lock Verification
- Centrifuge Verification
- Interface Verification
- Weight Scale Verification
- Leak Detector Verification
- Temperature Sensor Verification
- Cuff Verification
- USB Verification

- Power Verification
- Battery Verification
- Simulated AmiCORE Procedure

Section 3.2 Instrument Unpacking and Assembly

This procedure provides instructions for installation of the AmiCORE Apheresis System.

Note: The following procedure must be performed by authorized service personnel.

Note: Consult the AmiCORE Administrator's Guide for instructions on how to configure the Administrative Settings of the instrument.

Caution: When working inside the AmiCORE Apheresis System, service personnel must observe electrical safety precautions.

Caution: Before touching any printed circuit board (PCBA), service personnel should always touch chassis ground first. Follow proper anti-static procedures by implementing an anti-static wrist band.

Caution: Avoid wearing loose clothing and/or jewelry when servicing the instrument.

Caution: Ensure the AmiCORE is in a stable and level configuration which limits instrument movement.

Warning: Ensure caution around live mains power feed lines or internal access points

Unboxing

Caution: The AmiCORE Apheresis System weighs approximately 111 kg.

1. Open the AmiCORE Apheresis System shipping container and remove the packing foam.

2. Place the shipping container ramp in front of the AmiCORE Apheresis System.

3. Remove the shipping straps.

4. Unlock the AmiCORE wheels.

5. Carefully push the AmiCORE down the ramp.

6. Open the AmiCORE centrifuge door. Remove packaging on 0-omega arm and packaging between centrifuge and centrifuge compartment. Open centrifuge and remove any packaging. Ensure spool is latched into spool holder. Close and latch top plate. Close the centrifuge door.

7. Remove additional items from the shipping container.

8. Verify the following items were contained within the shipping container:

Quantity	Description
1	AmiCORE Apheresis System
1	Power Cord
1	Manual CD (Operator's Manual, Administrator's Guide, Software Release Notes)
1	Pressure Cuff
2	Fuses
1	Air Filter

9. Connect the power cord to the AmiCORE Apheresis System in the lower rear of the device. Plug the power cord into a dedicated power outlet.

10. Connect the cuff assembly to the cuff port located in the rear of the device.

11. Rotate each solution pole until they detent into position. Do not force rotation past detents.

12. For AmiCORE movement over flat surfaces, refer to AmiCORE Operator's Manual, Chapter 6. If the instrument has been moved

or relocated at any time after installation completion, weight scale accuracy checks must be performed to ensure scale accuracy.

Section 3.3 Visual Inspection

Perform a visual inspection of the AmiCORE Apheresis System.

Verify the following:

- All accessories received with instrument (See Instrument Unpacking and Assembly Section 3.1).
- Both scale hooks on left side of instrument hanging freely.
- Six pump assemblies with pump heads in place.
- No visible chips or cracks in AmiCORE Apheresis System casing.
- No missing parts.
- No physical damage.

Contact Fresenius Kabi customer service if any issues are found during the visual inspection.

Section 3.4 Power Up System Test

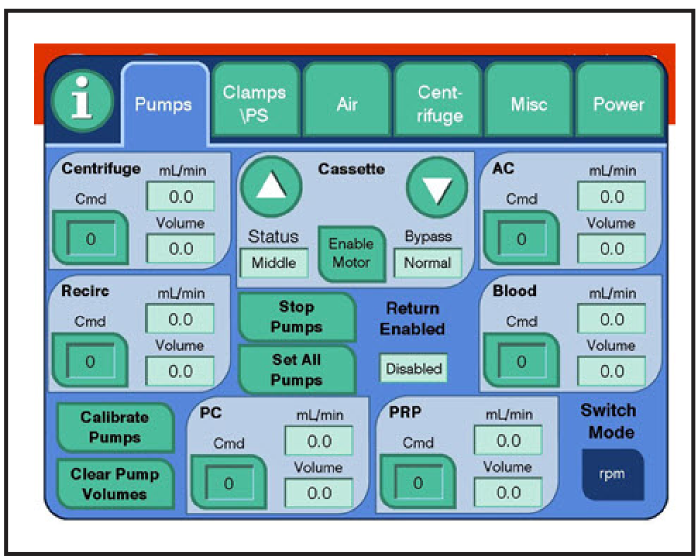
1. Switch the main power switch, located in the rear lower left hand side of the device, to the ON (up) position.
2. Switch the daily power switch, located in the rear of the display assembly, to the ON (up) position.
3. The AmiCORE Apheresis System will perform power up initialization tests. Verify that there is no “secure shell enabled” red banner on the STOP Button Test Screen.
4. When prompted, press the STOP *button*.
5. Verify the speaker plays an alert signal and touch the *check button*.

6. Verify the power up completes with no alerts and the AmiCORE Apheresis System displays the following options on screen: “Start New Procedure” and “View Last Donation Procedure Summary”.

Section 3.5 Enable Service Mode

Warning: A donor should not be connected to the instrument while in Service Mode.

1. Touch the *information button* (*I-button*) located in the upper left-hand corner of the display screen.
2. Verify the *instrument settings tab* is displayed.
3. Touch the *Service Mode button*. Touch the *on screen button*.
4. Enter the service password when prompted. The default password is “PASSWORD2”.
5. The service menu screens are now available. To exit Service Mode, power cycle the instrument.



Section 3.6 Pump Verification

1. Enable Service Mode (See Section 3.4).
2. Touch the *pumps tab*.
3. Touch the *cmd button* for the centrifuge pump.
4. Using the keypad, enter a commanded speed of 50. Touch the *check button*.
5. Verify the centrifuge pump is rotating in the clockwise direction.
6. Enter a commanded speed of 150. Verify the centrifuge pump accelerates.
7. Enter a commanded speed of 0. Verify the centrifuge pump stops.
8. Enter a commanded speed of -50. Verify the centrifuge pump is rotating in the counter clockwise direction.
9. Enter a commanded speed of -100. Verify the centrifuge pump accelerates.
10. Enter a commanded speed of 0. Verify the centrifuge pump stops.
11. Touch the *cmd button* for the recirc pump.
12. Using the keypad, enter a commanded speed of 20. Touch the *check button*.
13. Verify the recirc pump is rotating in the clockwise direction.
14. Enter a commanded speed of 45. Verify the recirc pump accelerates.
15. Enter a commanded speed of 0. Verify the recirc pump stops.
16. Enter a commanded speed of -20. Verify the recirc pump is rotating in the counter clockwise direction.
17. Enter a commanded speed of -45. Verify the recirc pump accelerates.
18. Enter a commanded speed of 0.
19. Repeat Steps 4 – 10 for the PC pump.
20. Repeat Steps 12 – 18 for the PRP pump.

21. Repeat Steps 4 – 10 for the Blood pump.

22. Repeat Steps 12 – 18 for the AC pump.

23. Touch the *set all pumps button*. Enter a commanded speed of 35.

24. Verify all six pumps are spinning in the clockwise direction.

25. Touch the *stop pumps button*. Verify all pumps stop.

Section 3.7 Cassette Motor Verification

1. Enable Service Mode (See Section 3.4).

2. Touch the *pumps tab*.

3. Touch the *cassette enabling button* to read disable motor.

4. Touch the cassette *down arrow*.

5. Verify the cassette plate lowers.

6. Verify the cassette *status field* displays loaded.

7. Touch the cassette *up arrow*.

8. Verify the cassette plate raises to the upright position.

9. Verify the cassette *status field* displays unloaded.

10. Touch the *cassette enabling button* to read enable motor.

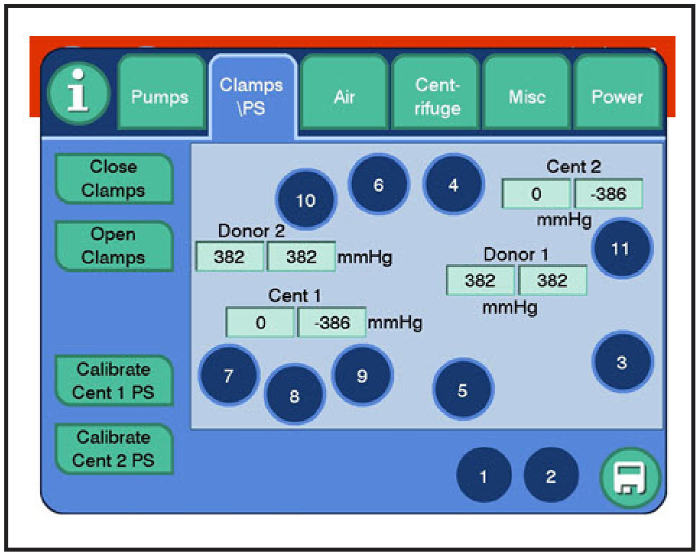
11. Touch the cassette *down arrow*.

12. Verify the centrifuge plate does not lower.

Section 3.8 Clamp Verification

1. Enable Service Mode (See Section 3.4).

2. Touch the *clamps\PS tab*.



3. Touch the *open clamps button*. Note: The clamp #1 and #2 buttons are green when clamps are open, and blue when closed.

4. Visually verify that both donor clamps, identified as “1” and “2” on the service *clamps\PS tab*, are open.

5. Touch the *1 clamp button*.

6. Visually verify that donor clamp 1 closes.

7. Touch the *2 clamp button*.

8. Visually verify that donor clamp 2 closes.

9. Touch the *open clamps button*.

10. Visually verify that both donor clamps, identified as “1” and “2” on the service *clamps\PS tab*, are open.

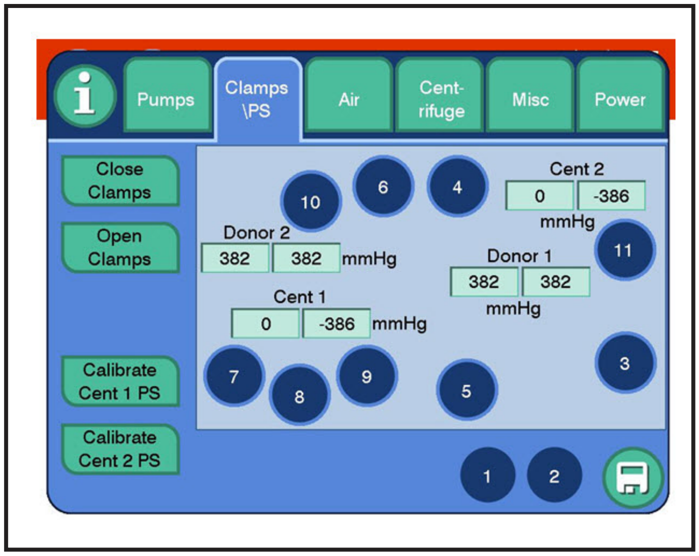
11. Touch the *close clamps button*.

12. Visually verify that both donor clamps, identified as “1” an “2” on the service *clamps\PS tab*, are closed.

Section 3.9 Valve Verification

1. Enable Service Mode (See Section 3.4).

2. Touch the *clamps\PS tab*.



3. Touch the *open clamps button*.

4. Visually verify that valves 3, 4, 5, 6, 7, 8, 9, 10, and 11 are all open (down position).

5. Touch the *valve 3 button*. Verify valve 3 closes.

6. Touch the valve *3 button*. Verify valve 3 opens.

7. Touch the *valve 4 button*. Verify valve 4 closes.

8. Touch the *valve 4 button*. Verify valve 4 opens.

9. Touch the *valve 5 button*. Verify valve 5 closes.

10. Touch the *valve 5 button*. Verify valve 5 opens.

11. Touch the *valve 6 button*. Verify valve 6 closes.

12. Touch the *valve 6 button*. Verify valve 6 opens.

13. Touch the *valve 7 button*. Verify valve 7 closes.

14. Touch the *valve 7 button*. Verify valve 7 opens.

15. Touch the *valve 8 butto*n. Verify valve 8 closes.

16. Touch the *valve 8 button*. Verify valve 8 opens.

17. Touch the *valve 9 button*. Verify valve 9 closes.

18. Touch the *valve 9 button*. Verify valve 9 opens.

19. Touch the *valve 10 button*. Verify valve 10 closes.

20. Touch the *valve 10 button*. Verify valve 10 opens.

21. Touch the *valve 11 button*. Verify valve 11 closes.

22. Touch the *valve 11 button*. Verify valve 11 opens.

23. Touch the *close clamps button*.

24. Visually verify valves 3, 4, 5, 6, 7, 8, 9, 10, and 11 close.

Section 3.10 Pressure Sensor Verification

Note: This verification requires an AmiCORE cassette, digital pressure meter, syringe, Y-connector tubing, and two hemostats.

1. Enable Service Mode (See Section 3.4).

2. Touch the *pumps tab*.

3. Install the AmiCORE cassette on the cassette plate.

4. Ensure the *cassette button* reads disable motor. If the button reads enable motor, touch the button.

5. Set all pumps to -30 rpm.

6. Touch the cassette *down arrow button*.

7. Stop all pumps. Reverse all pumps at 30 rpm, if necessary, to fully load pump tubing within pump raceway. Reverse again at -30 rpm if necessary.

8. Connect the digital pressure meter to the Y-tubing.

9. Connect the syringe to one of the Y-tubing tubes.

10. Touch the *clamps\PS tab*.

11. Touch the *open clamps button*.

Donor Pressure Sensor #1

12. Attach the Y-tube to the first tube to the right of the PRP pump.

13. Place a hemostat on the fourth tubing to the right of the PRP pump

14. Apply a pressure with the syringe of approximately 100 mmHg on the digital pressure meter.

15. Verify that both sensor readings for donor 1 display the value on the digital pressure meter ± 50 mmHg

16. Apply a pressure with the syringe of approximately 450 mmHg on the digital pressure meter.

17. Verify that both sensor readings for donor 1 display the value on the digital pressure meter ± 50 mmHg.

18. Apply a pressure with the syringe of approximately -100 mmHg on the digital pressure meter.

19. Verify that both sensor readings for donor 1 display the value on the digital pressure meter ± 50 mmHg.

20. Apply a pressure with the syringe of -250 mmHg.

21. Verify that both sensor readings for donor 1 display the value on the digital pressure meter ± 50 mmHg.

22. Remove the Y-tubing from the cassette.

Donor Pressure Sensor #2

23. Attach the Y-tube to the third tubing over from the top left corner of the cassette.

24. Touch the *valve #5 button* to close valve #5.

25. Apply a pressure with the syringe of approximately 100 mmHg on the digital pressure meter.

26. Verify that both sensor readings for donor 2 display the value on the digital pressure meter ± 50 mmHg.

27. Apply a pressure with the syringe of approximately 450 mmHg on the digital pressure meter.

28. Verify that both sensor readings for donor 2 display the value on the digital pressure meter ± 50 mmHg.

29. Apply a pressure with the syringe of approximately -100 mmHg.

30. Verify that both sensor readings for donor 2 display the value on the digital pressure meter ± 50 mmHg.

31. Apply a pressure with the syringe of approximately -250 mmHg.

32. Verify that both sensor readings for donor 2 display the value on the digital pressure meter ± 50 mmHg.

33. Remove the Y-tubing from the cassette.

Centrifuge Pressure Sensor #1

34. Attach the Y-tube to the sixth tube over from the top left corner of the cassette.

35. Apply a pressure with the syringe of approximately 200 mmHg on the digital pressure meter.

36. Verify that both sensor readings for Cent 1 display the value on the digital pressure meter ± 200 mmHg.

37. Apply a pressure with the syringe of approximately 500 mmHg on the digital pressure meter.

38. Verify that both sensor readings for Cent 1 display the value on the digital pressure meter ± 100 mmHg.

39. Apply a pressure with the syringe of approximately 750 mmHg on the digital pressure meter.

40. Verify that both sensor readings for Cent 2 display the value on the digital pressure meter ± 100mmHg.

41. Remove the Y-tubing from the cassette.

Centrifuge Pressure Sensor #2

42. Attach the Y-tube to the ninth tube from the top left of the cassette.

43. Apply a pressure with the syringe of approximately 100 mmHg on the digital pressure meter.

44. Verify that both sensor readings for Cent 2 display the value on the digital pressure meter ± 200 mmHg.

45. Apply a pressure with the syringe of approximately 500 mmHg on the digital pressure meter.

46. Verify that both sensor readings for Cent 2 display the value on the digital pressure meter $\pm$ 100 mmHg.

47. Apply a pressure with the syringe of approximately 750 mmHg on the digital pressure meter.

48. Verify that both sensor readings for Cent 2 display the value on the digital pressure meter $\pm$ 100 mmHg.

49. Remove the Y-tubing from the cassette.

50. Set all pumps to - 30 RPM.

51. Select the up option for the cassette.

52. Stop all pumps.

53. Remove the test cassette.

Section 3.11 Air Detector Verification

Note: This verification requires a fluid-filled tubing.

1. Enable Service Mode (See Section 3.4).

2. Ensure nothing is placed within the air detector.

3. Touch the *clear air latch clear button*.

4. Touch the *clear safety air latch clear button*.

5. Verify the air message, air pulse message, safety air message, and safety air pulse message output fields display *air*.

6. Place the fluid-filled tubing in the air detector.

7. Touch the *clear air latch clear button*.

8. Touch the *clear safety air latch clear button*.

9. Verify the air message, air pulse message, safety air message, and safety air pulse message output fields display *liquid*.

10. Remove the fluid-filled tubing from the air detector.

11. Verify the air message, air pulse message, safety air message, and safety air pulse message output fields display *air*.

12. Place the fluid-filled tubing in the air detector.

13. Touch the *clear air latch clear butto*n.

14. Touch the *clear safety air latch clear butto*n.

15. Verify the air message, air pulse message, safety air message, and safety air pulse message output fields display *liquid*.

16. Touch the *air test message normal button* to test the air detector.

17. Verify the air message, air pulse message, safety air message, and safety air pulse message output fields display *air*.

18. Touch the *air test message test butto*n to return to normal mode on the air detector.

19. Touch the *clear air latch clear butto*n.

20. Touch the *clear safety latch clear butto*n.

21. Touch the arm for return *disarm button*.

22. Verify the armed for return output field displays *arm*.

23. Verify the returned enabled output field displays *enabled*. If it is *disabled*, ensure the fluid-filled tubing is in place and clear both latches until all output fields display *fluid*.

24. Touch the *pumps tab*.

25. Verify the return enabled output field displays *enabled*.

26. Select the *cmd button* for the blood pump.

27. Set the blood pump speed to -100.

28. Verify the blood pump is rotating counterclockwise.

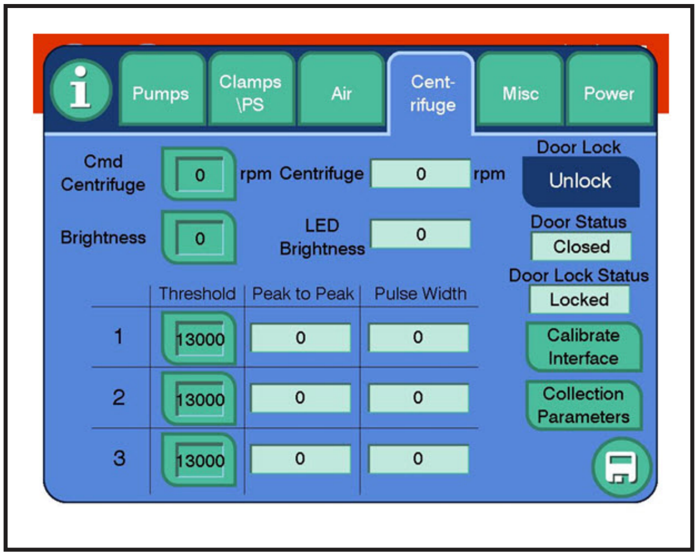
29. Remove the tubing from the air detector.

30. Verify that return enabled has switched to *disabled* and that the blood pump stops. The system will also raise an alert (10005 Pump Speed Error) after a few seconds. The alert is expected.

Section 3.12 Door Lock Verification

1. Enable Service Mode (See Section 3.4).

2. Touch the *centrifuge tab*.



3. Verify the *door lock button* displays lock. If the *door lock button* displays *unlock*, touch the button to unlock the door.

4. Verify the door lock status output field displays *unlocked*.

5. Verify the centrifuge compartment door is unlocked by opening it.

6. Verify the door status output field displays *open*.

7. Physically close the centrifuge compartment door.

8. Verify the door status output field displays *closed*.

9. Touch the *door lock button*. Verify it displays *unlock*.

10. Verify the door door status output field displays *locked*.

11. Physically verify that the door is locked and cannot be opened.

12. Touch the *door lock button*.

13. Physically verify that the door is unlocked and can be opened.

Section 3.13 Centrifuge Verification

Note: This verification requires an AmiCORE disposable kit umbilicus.

1. Enable Service Mode (See Section 3.4).

2. Touch the *centrifuge tab*.

3. If the *door lock button* displays *unlock*, touch the *door lock button*.

4. Verify the *door lock button* displays *lock*.

5. Open the centrifuge and install the umbilicus.

6. Close the centrifuge door.

7. Touch the *door lock button*. Verify the *door lock button* displays *unlock*. Verify the door lock status displays *locked*.

8. Touch the *cmd centrifuge button*.

9. Enter a commanded centrifuge speed of 1640.

10. Verify the centrifuge output field increases from 0 to 1640 rpm ± 164 rpm at steady state.

11. Touch the *cmd centrifuge button*.

12. Enter a commanded centrifuge speed of *0*.

13. Verify the centrifuge output field decreases to 0 rpm.

14. Touch the *door lock button*

15. Remove the umbilicus from the centrifuge.

Section 3.14 Interface Verification

Note: The verification requires a fluid filled umbilicus and pack assembly with no air bubbles in the pack.

1. Enable Service Mode (See Section 3.4).
2. Install the fluid-filled umbilicus and pack in the centrifuge. (Note: This may be accomplished by performing a simulated procedure (See Section 3.21). During collection mode, pause the procedure. Hemostat the lines going into the centrifuge. End the procedure and do not remove the umbilicus/pack.)
3. Close the door.
4. Touch the *centrifuge tab*.
5. Touch the *door lock button* to lock the centrifuge door.
6. Touch the *cmd centrifuge button* and enter 1640.
7. Verify that the thresholds are all set to 13000.
8. Once the centrifuge has reached steady state at the commanded speed, select the *calibrate interface button*.
9. Verify the calibrate interface procedure completes with no alerts.

Section 3.15 Weight Scale Verification

Note: This verification requires a 1 kg weight.

1. Enable Service Mode (See Section 3.4).
2. Touch the *misc tab*.
3. Touch the -> *0 <- button* for weight scale 1.
4. Verify the *net weight output field* displays 0 g $\pm$ 5 g.
5. Place the 1 kg weight on the AmiCORE Apheresis System scale hook.

6. Verify the *net weight output field* displays 1000g $\pm$ 5 g.
7. Remove the 1000 g weight.
8. Touch the -> *0* <- *button* for weight scale 2.
9. Repeat Steps 4 – 7 for weight scale 2.

Section 3.16 Leak Detector Verification

1. Enable Service Mode (See Section 3.4).
2. Touch the *misc tab*.
3. Verify the *leak sensor 1 output field* displays a value between 5139 and 16958.
4. Verify the *leak sensor 2 output field* displays a value between 5139 and 16958.
5. Open the centrifuge door.
6. Place a damp cloth across the two leak detector strips.
7. Verify the *leak sensor 1 output field* displays a value greater than 16958.
8. Verify the *leak sensor 2 output field* displays a value greater than 16958.
9. Dry off the leak sensor strips.
10. Close the centrifuge door.

Section 3.17 Temperature Sensor Verification

1. Enable Service Mode (refer Section 3.4).
2. Touch the *misc tab*.

3. Verify that the readings of *temperature sensor 1 output field* and *temperature sensor 2 output field* are within 4.6 degrees Celsius of each other.

Section 3.18 Cuff Verification

1. Enable Service Mode (See Section 3.4).
2. Ensure the AmiCORE Apheresis System pressure cuff is securely fastened to the rear of the device.
3. Touch the *misc tab*.
4. Verify the *cuff pressure output field* displays less than 15 mmHg.
5. Touch the *target cuff pressure button*.
6. Enter a value of 50 mmHg.
7. Verify the pressure cuff inflates.
8. Verify the *cuff pressure output field* displays 50 mmHg ± 12 mmHg.
9. Touch the *target cuff pressure button*.
10. Enter a value of 0 mmHg.
11. Verify the pressure cuff deflates.
12. Verify the *cuff pressure output field* displays less than 15 mmHg.
13. Touch the *target cuff pressure button*.
14. Enter a value of 100 mmHg.
15. Verify the cuff inflates.
16. Verify the *cuff pressure output field* displays 100 mmHg ± 12 mmHg.
17. Once steady state has been reached, verify the compressor does not actuate more than once per minute.
18. Set the target cuff pressure to 0 mmHg.

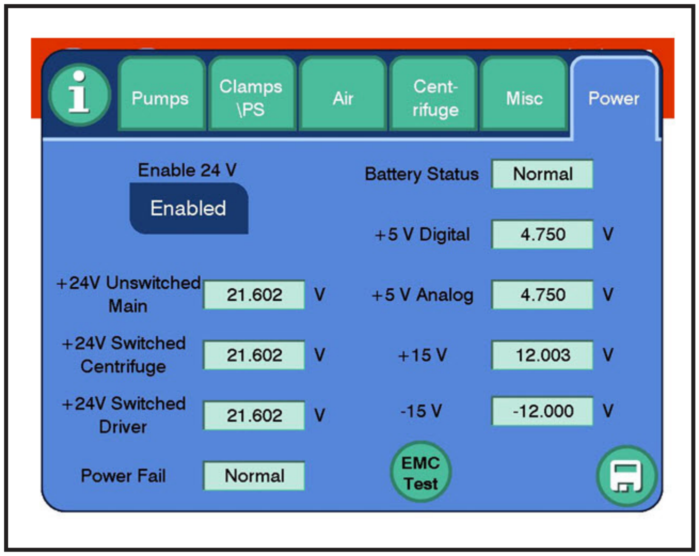
Section 3.19 USB Verification

Note: This verification requires an AmiCORE USB drive and bar code scanner (optional).

1. Enable Service Mode (See Section 3.4)
2. Touch the *misc tab*.
3. Insert the USB drive into the left USB port located in the rear of the AmiCORE Apheresis System.
4. Verify the *USB status output field* displays Valid USB.
5. Verify the USB class output field displays *3* or *8*.
6. Remove the USB drive from the left USB port and place it in the right USB port located in the rear of the AmiCORE Apheresis System.
7. Verify the *USB status output field* displays Valid USB.
8. Verify the *USB class output field* displays 3 or 8.
9. Remove the USB drive from the USB port.
10. Optional: For use of bar code scanner, connect the AmiCORE bar code scanner to a USB port. Verify the *USB status output field* displays Valid USB. Verify the USB Class output field displays 8.

Section 3.20 Power Verification

1. Enable Service Mode (See Section 3.4).
2. Touch the *power tab*.



3. Touch the *enable 24V butto*n to display disabled.

4. Verify the *+24V switched centrifuge output field* displays less than 5.02 V.

5. Verify the *+24V switched driver output field* displays less than 5.02 V.

6. Verify the *+24V unswitched main output field* displays a value between 21.6V and 25.2 V.

7. Touch the *enable 24V button* to display enabled.

8. Verify the *+24V unswitched main output field* displays a value between 21.6V and 25.2 V.

9. Verify the *+24V switched centrifuge output fiel*d displays a value between 21.6 V and 25.2 V.

10. Verify the +24V *switched driver output field* displays a value between 21.6 V and 25.2 V.

11. Verify the *battery status output field* displays normal. (Note: If battery level is low, this field may display fault.)

12. Verify the *+5 V digital output field* displays a value between 4.75 V and 5.35 V.

13. Verify the *+5 V analog output field* displays a value between 4.75 V and 5.35 V.

14. Verify the *+15 V output field* displays a value between 12 V and 15.75 V.

15. Verify the *-15V output field* displays a value between -12 V and -15.75 V.

16. Verify the *power fail output field* displays normal.

17. Remove AC power cord.

18. Verify the *power fail output field* displays fault.

19. Replace the AC power cord. An alert may occur after replacing the power cord; acknowledge the alert.

Section 3.21 Battery Verification

1. Ensure the AmiCORE Apheresis System is in normal operations mode. If in Service Mode, exit Service Mode and cycle power.

2. From the main menu, touch the *start new procedure button*.

3. Enter Jane123 for operator ID.

4. Carefully pull the AmiCORE Apheresis System power cord from the power socket.

5. Carefully replace the AmiCORE Apheresis System power cord into the power socket.

6. Verify the following alert appears: 10002.

7. Acknowledge the alert.

8. Verify the operator ID displays Jane123.

Section 3.22 Simulated AmiCORE Procedure

Note: The simulated AmiCORE procedure requires an unused AmiCORE instrument disposable kit and two solution bags with at least 500 mL of saline in each bag. Additionally, a container of water is required.

1. Ensure the AmiCORE Apheresis System is in normal operations mode. If the instrument is in Service Mode, exit Service Mode and cycle instrument power.

2. Touch the *information button*, select *demo mode*. This option places the AmiCORE in demo mode which enables the ability to run saline.

3. Touch the *start new procedure button*.

4. Enter Jane123 for the operator ID.

5. Touch the *check button* on the *enter kit information tab*.

6. Follow the on screen instructions for loading the kit. Hang the bags and load the cassette. Touch the *next arrow*.

7. Follow the on screen instructions for loading the kit. Load the centrifuge pack and route the tubing. Touch the *check button*.

8. Touch the *check button* for the *enter saline information tab*.

9. Follow the on screen instructions for loading the saline bag. Touch the *confirm saline bag button* and enter the volume of the saline bag. Touch the *check button*.

10. Touch the *check button* for the *enter anticoagulant information tab*.

11. Follow the on screen instructions for loading second container of Saline. Touch the *confirm AC bag volume button* and enter the volume of the saline bag. Touch the *check button*.

12. Enter 12345 for the procedure ID. Touch the *check button*.

13. Enter 6789 for the donor ID. Touch the *check button*.

14. Select the *male icon*. Touch the *check button*.

15. Enter 133 cm for the *enter height tab*. Touch the *check button*.

16. Enter 240 kg for the *enter weight tab*. Touch the *check button*.

17. Enter 42% for the *enter hematocrit tab*. Touch the *check button*.

18. Enter 295 for the *enter pre-count tab*. Touch the *check button*.

19. Touch 3.0 for the *select product yield tab*. Touch the *check button*.

20. Touch the *check button* on the Parameter Entry Summary Screen.

21. Place the donor fistula in a container of water and ensure the donor line roller clamp is closed.

22. The AmiCORE Apheresis System will automatically prime the kit with saline. After approximately five minutes, the *start button* will enable. Open the roller clamp, and touch the *start button*.

23. During a draw cycle, indicated by the arrows pointing away from the donor icon, close the clamp on the donor line. Verify an inlet line alert occurs.

24. Open the clamp on the donor line. Acknowledge the alert. Verify the procedure continues.

25. During a return cycle, indicated by the arrows pointing towards the donor icon, remove the line from the air detector. Verify an air detected alert occurs.

26. Place the line back into the air detector. Select the option to purge air.

27. Acknowledge the alert. Verify the procedure continues.

28. During a return cycle, close the clamp on the donor line. Verify a return line occlusion alert occurs.

29. Open the clamp on the donor line. Acknowledge the alert. Verify the procedure continues.

30. Press the STOP *button*. Choose the option to end the procedure.

31. Follow the on-screen instructions to end the procedure and remove the kit.

32. Power the AmiCORE Apheresis instrument OFF.

Chapter 4 - Preventative Maintenance

Introduction

The preventative maintenance procedure is a series of instrument checks performed on an annual basis to ensure the AmiCORE Apheresis System is operating as expected. The preventative maintenance procedure may also be used, if/when necessary, to clean or replace parts deemed to be replaced annually.

This chapter contains information pertaining to the frequency, tests, and documentation for the AmiCORE Apheresis System preventative maintenance procedure.

The preventative maintenance procedure is to be performed by trained service personnel only.

Section 4.1 Preventative Maintenance Frequency

The preventative maintenance procedure is performed at a minimum of once per calendar year with a tolerance of $\pm$ 30 days.

Section 4.2 Parts Replacement Preventative Maintenance

The AmiCORE Apheresis System requires the proactive replacement of the following parts:

- Centrifuge Gimbal Ring (See Section 5.32)
- Air Filter (See Section 5.36)
- Donor Pressure Sensor #1 O-ring (See Section 5.33)
- Donor Pressure Sensor #1 Cover (See Section 5.34)
- Donor Pressure Sensor #2 O-ring (See Section 5.33)
- Donor Pressure Sensor #2 Cover (See Section 5.34)

Any parts identified as faulty during the execution of the preventative maintenance procedure should be replaced and tested per appropriate replacement procedure.

Any parts replaced or adjustments made during the preventative maintenance Procedure should be noted on the *comments field* of the Installation and Preventative Maintenance Checklist.

Section 4.3 Preventative Maintenance Inspect and Cleaning

Inspect the following areas for signs of wear. Replace parts as necessary:

- Pump Heads
- Pressure Cuff

Clean the following parts and areas. Refer to the AmiCORE Operator's Manual for approved cleaning solutions and disinfecting instructions.

- All six pump heads and surrounding pump housing
- Centrifuge compartment
- Centrifuge spool holder, spool, and spool holder window
- Interface window
- Cassette nest (below cassette plate)

Inspect the following areas to ensure they do not tear or cut corresponding disposable kit components:

- Pump Heads
- Solenoid Valves
- Donor Clamps
- Centrifuge Rotor
- Spool
- Spool Holder
- Centrifuge Door

Section 4.4 Preventative Maintenance Test Procedure

The Preventative Maintenance procedure is defined by the Installation and Preventative Maintenance Checklist. The checklist is used with the AmiCORE Apheresis System Service Manual to perform the preventative maintenance procedure. The procedure includes the following tests:

- Visual Inspection
- Pump Verification
- Cassette Motor Verification
- Clamp Verification
- Valve Verification
- Pressure Sensor Verification
- Air Detector Verification
- Door Lock Verification
- Centrifuge Verification
- Interface Verification
- Weight Scale Verification
- Leak Detector Verification
- Temperature Sensor Verification
- Pressure Cuff Verification
- USB Verification
- Power Verification
- Battery Verification
- Simulated Procedure

In addition to the tests listed, the preventative maintenance includes proactive parts replacement outlined in Section 4.2, and the inspection and cleaning outlined in Section 4.3.

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Chapter 5 - Parts Removal

Introduction

This section describes the steps and tools required to replace AmiCORE apheresis instrument parts.

Caution: When working inside the AmiCORE instrument, service personnel must observe electrical safety precautions.

Caution: The AmiCORE Apheresis System must be disconnected from external power sources when performing parts removal.

Caution: Replace all shields, covers, screws, and gaskets in their exact locations; failure to do so may result in increased emissions or decreased immunity of the instrument.

Caution: No donor should be connected to instrument while in Service Mode.

Caution: Avoid touching the Touch Screen with sharp objects.

Caution: Follow proper protective measures by using personal protective equipment where appropriate.

Note: Follow all regional and/or environmental regulations when disposing of any removed AmiCORE Apheresis System parts.

Note: An anti-static wrist strap should be worn when performing the parts removal or replacement of any electrical parts (i.e., circuit boards). Observe proper ESD precautions.

Note: While the focus of this chapter is parts removal, it may contain additional specific information on parts installation, when applicable.

Section 5.1 Tool Listing

The following tools are required for executing the parts removal instructions in this section:

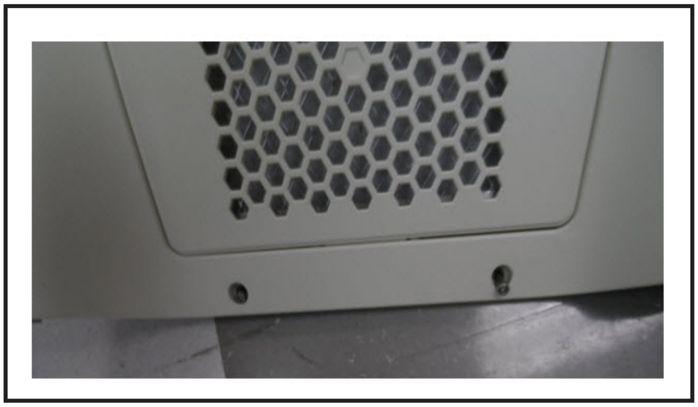
Tool	Size	Notes	Special Tool
Hex Wrench	2.5 mm	T-Handle	
Hex Wrench	3 mm	T-Handle	
Hex Wrench	4 mm	T-Handle	
Hex Wrench	4 mm	Right Angle	
Hex Wrench	5 mm	T-Handle	
Hex Wrench	5 mm	Right angle	
Hex Wrench	6 mm	T-Handle	
PHILLIPS Screwdriver	Medium		
Standard Screwdriver	Small		
Needle Nose Pliers			
Wire cutters			
Adjustable Crescent Wrench			
Nut Driver	18 mm	Deep well	
Nut Driver	7 mm		
Pressure Sensor Removal			X
Centrifuge Nut Removal			X
Clamp Removal			X

Section 5.2 Front Panel Removal

Tools Required:

- 4 mm hex wrench T-Handle

1. Locate the two screws along the lower edge of the front panel. These two screws hold the front panel in place. (Note for replacement: The front panel is keyed along the top edge, no fasteners required).



2. Fully loosen both of the captive screws along the lower edge of the front panel.

3. Carefully lower the front panel and remove.

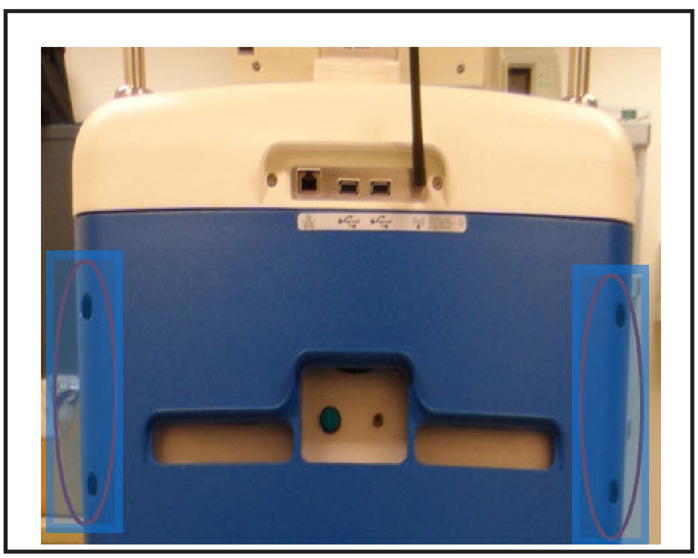
Section 5.3 Rear Top Panel Removal

Tools Required:

- 4 mm hex wrench T-Handle

1. Fully loosen the two screws along the left edge of the rear top panel.

2. Fully loosen the two captive screws along the right edge of the rear top panel.



3. Carefully remove the rear top panel. Place in a secure area.

Section 5.4 Rear Bottom Panel Removal

Tools Required:

- 4 mm hex wrench T-Handle

1. Remove the rear top panel (See Section 5.3).

2. Fully loosen the four screws along the left edge of the rear lower panel.

3. Fully loosen the four screws along the right edge of the rear lower panel.

4. Carefully remove the two lowest screws and corresponding washers and then remove the rear lower panel and place in a secure area.

Section 5.5 Driver Board Removal

Tools Required:

- 7 mm nut driver

Caution: An anti-static grounding strap must be worn while performing these steps.

Note: This procedure involves the disconnection of several connectors. It is suggested that connectors are temporarily labeled with the corresponding connector number to aid in installation of same/replacement driver board.

1. Remove the front panel (See Section 5.2).
2. Carefully unplug the four connectors along the left edge of the driver board (clamps connector and three pump connectors).
3. Carefully unplug the power supply connector from the lower edge of the driver board.
4. Carefully unplug the five connectors along the right edge of the driver board (clamps connector, three pump connectors, and main connector).
5. Carefully remove the two remaining connectors along the top edge of the driver board (cassette bypass button and cassette motor).
6. Remove the ground wire along the lower edge of the driver board with a 7mm nut driver.
7. Carefully pull the board off of the eight press fit connectors.

Section 5.6 Main Board Removal

Caution: An anti-static grounding strap must be worn while performing these steps.

Note: This procedure involves the disconnection of several connectors. It is suggested that connectors are temporarily labeled with the corresponding connector number to aid in installation of same/replacement driver board.

1. Remove the rear top panel (See Section 5.3).
2. Remove the rear lower panel (See Section 5.4).
3. Carefully unplug the connectors along the left edge of the main board (SBC, driver, air detector, stop button, power).
4. Carefully unplug the connector from the bottom edge of the main board (compressor).
5. Carefully unplug the connectors along the right edge of the main board (PS1, PS board, PS4, weight scale board).
6. Carefully unplug the connector from the upper edge of the main board (interface controller).
7. Carefully remove the compressor tubing from connector U23.
8. Carefully remove the main driver board from the four press fit connectors.

Section 5.7 Interface Controller Board Removal

Caution: An anti-static grounding strap must be worn while performing these steps.

Note: This procedure involves the disconnection of several connectors. It is suggested that connectors are temporarily labeled with the corresponding connector number to aid in installation of the same/replacement driver board.

1. Remove the rear top panel (See Section 5.3).
2. Remove the rear lower panel (See Section 5.4).
3. Carefully unplug the connectors along the left edge of the interface controller board (door lock power and power).
4. Carefully unplug the connector from the lower edge of the interface controller board (main).

5. Carefully unplug the connectors from the right edge of the interface controller board. The fiber optic connections at PD1, PD2, and PD3 are threaded connections, carefully unscrew the connector and pull directly back to remove (three interface fiber optic connections and interface).

6. Carefully unplug the connectors from along the top of the interface controller board (cent board, door latch, door lock, and leak board).

7. Carefully remove the interface controller board from the press fit connectors

Section 5.8 Centrifuge Controller Board Removal

Caution: An anti-static grounding strap must be worn while performing these steps.

Note: This procedure involves the disconnection of several connectors. It is suggested that connectors are temporarily labeled with the corresponding connector number to aid in installation of the same/replacement driver board.

1. Remove the Rear Top Panel (See Section 5.3)

2. Remove the Rear Lower Panel (See Section 5.4).

3. Carefully remove the connectors along the top edge of the centrifuge controller board.

4. Carefully remove the large connector along the right edge of the centrifuge controller board (centrifuge).

5. Carefully remove the two connectors along the lower edge of the centrifuge controller board (power and interface).

6. Carefully remove the centrifuge controller board from the press fit connectors.

Section 5.9 Weight Scale Board Removal

Caution: An anti-static grounding strap must be worn while performing these steps.

Note: This procedure involves the disconnection of several connectors. It is suggested that connectors are temporarily labeled with the corresponding connector number to aid in installation of the same/replacement driver board.

1. Remove the rear top panel (See Section 5.3).

2. Remove the rear lower panel (See Section 5.4).

3. Carefully remove the two weight scale connectors from the left edge of the weight scale board.

4. Carefully remove the connector from the lower edge of the weight scale board (main).

5. Carefully remove the weight scale board from the press fit connectors.

Section 5.10 Leak Detector Board Removal

Tools Required:

- 2.5 mm hex wrench T-handle

Caution: An anti-static grounding strap must be worn while performing these steps.

Note: This procedure involves the disconnection of several connectors. It is suggested that connectors are temporarily labeled with the corresponding connector number to aid in installation of the same/replacement driver board.

1. Remove the rear top panel (See Section 5.3).

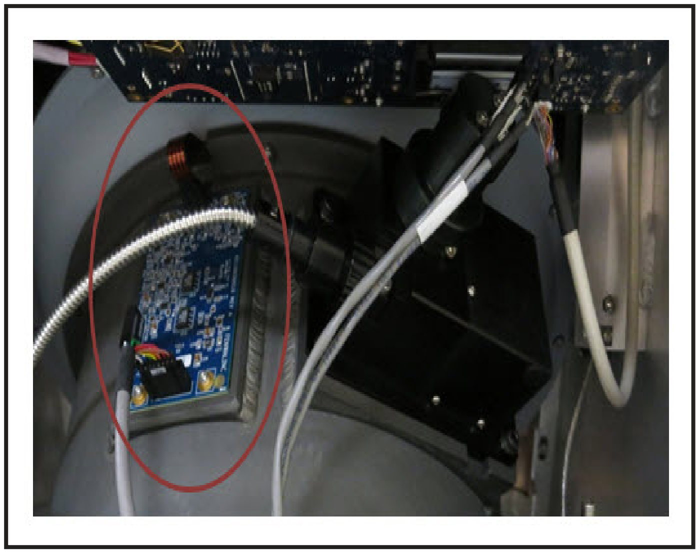
2. Carefully pull open the fan frame assembly panel (opens back and down).

3. Carefully remove the connector along the top edge of the leak board.

4. Carefully remove the connector along the lower edge of the leak board.

5. Remove the four screws from the leak detector board.

6. Carefully remove the leak detector board.



Section 5.11 Interface Detector Assembly Removal

Tools Required:

- Standard screwdriver
- 4 mm right angle hex wrench.
- Wire cutter

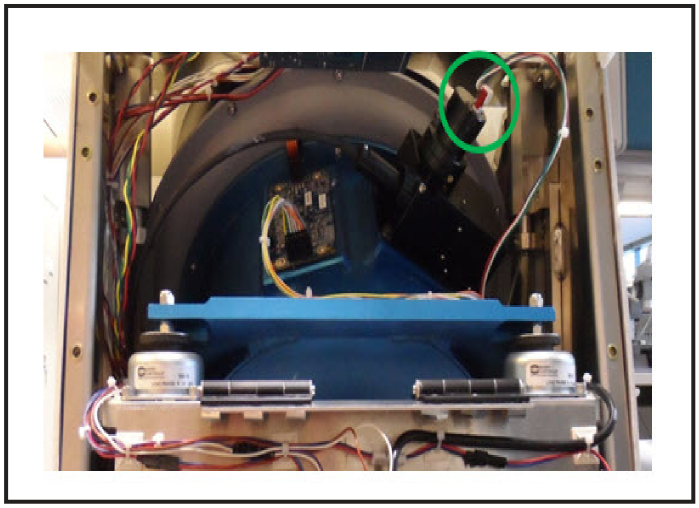
Caution: An anti-static grounding strap must be worn while performing these steps.

Note: This procedure involves the disconnection of several connectors. It is suggested that connectors are temporarily labeled with the corresponding connector number to aid in installation of same/replacement driver board.

1. Remove the rear top panel (See Section 5.3).

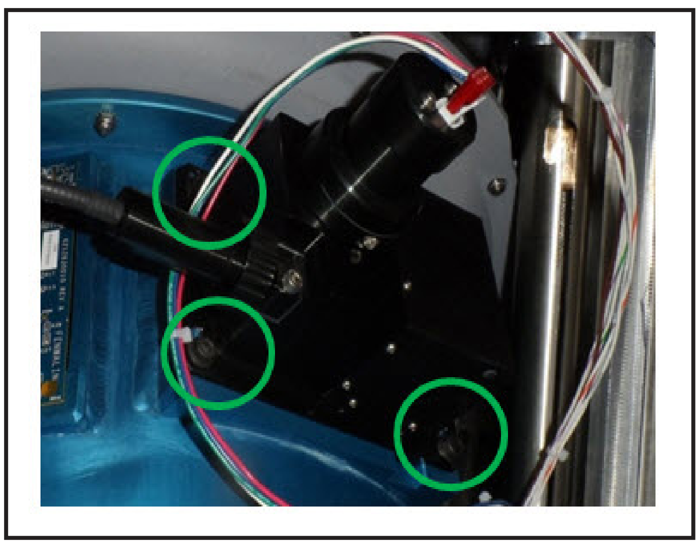
2. Carefully pull open the fan frame assembly forward and down.

3. Carefully remove the connector from the rear of the interface detector assembly.



4. Carefully remove the three interface optical connections from the interface board (PD1, PD2, PD3). These connectors are threaded, unscrew and pull directly back to remove. Clip the cable ties that the interface cable is threaded though.

5. Remove the four screws from interface detector assembly.



6. Carefully remove the interface detector assembly and cable.

Note: When reinstalling, apply LOCTITE 242 to screw threads. Tighten screws to 44 in-lbs.

Section 5.12 Pressure Sensor Board Removal

Tools Required:

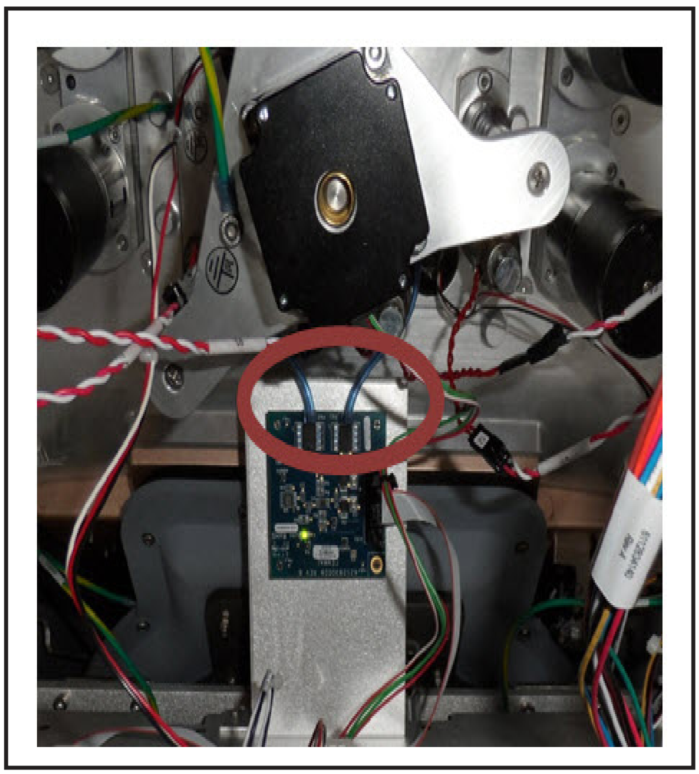
- Needle nose pliers

Caution: An anti-static grounding strap must be worn while performing these steps.

1. Remove the front panel (See Section 5.2).

2. Remove the connector to the pressure sensor board.

3. Using the needle nose pliers, carefully remove the two pressure sensor tubings from the pressure sensor board.



4. Carefully remove the pressure sensor board from the press fit connectors.

Note: If replacing the pressure sensor board, the suggested method for installation of the new board is to first remove the pressure sensor housings using the pressure sensor nut driver tool, replace the tubing and bushings, and then reinstall the pressure sensor housings. Once complete, install the pressure sensor board and reconnect the tubings.

Section 5.13 Pump Assembly Removal

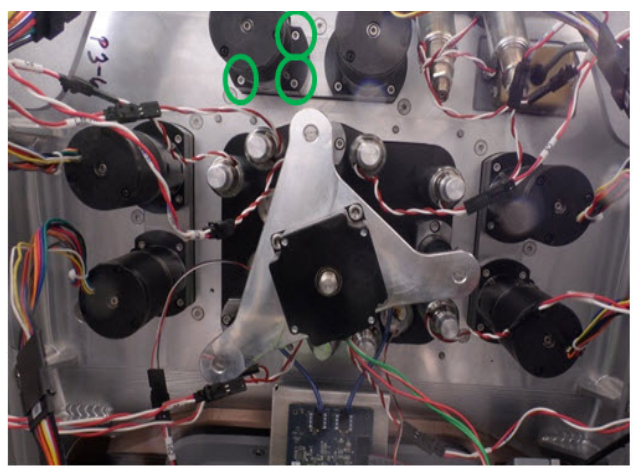
Tools Required:

- 2.5 mm hex wrench T-Handle (Long)

1. Remove the front panel (See Section 5.2).

2. Unplug the connector to the pump being removed from the main controller board.

3. Remove the four screws from the bottom of the pump.



5.1 Example of Pump Screw Location for PRP Pump

4. Carefully remove the pump assembly, taking care to not loosen the key that mates with flat side of pump motor.

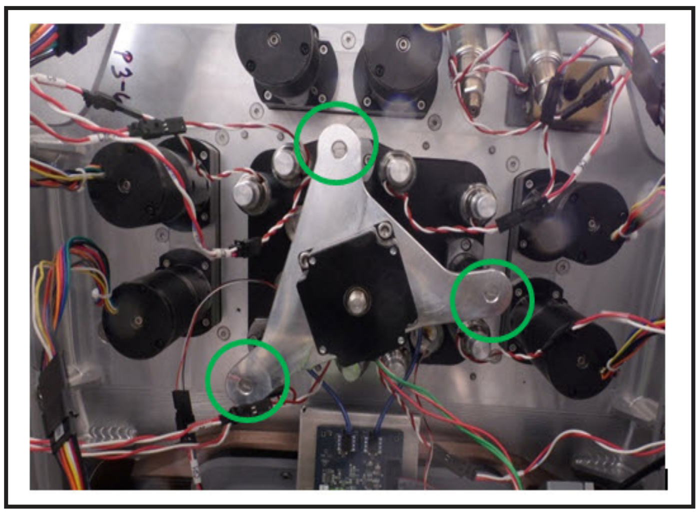
Note: When reinstalling, apply LOCTITE 242 to screw threads. Tighten screws to 26 in-lbs.

Section 5.14 Cassette Motor Removal

Tools Required:

- PHILLIPS screwdriver
- 4 mm hex wrench
- Adjustable wrench

1. Remove the front panel (See Section 5.2).
2. Remove the three screws from the cassette motor plate with the PHILLIPS screwdriver.



3. Remove the three screws on the top of the cassette plate (on top of device) and carefully remove the plate and the three plate poles.
4. Disconnect the motor cable from the board. Disconnect motor ground wire.
5. Carefully spin the cassette motor plate counter clockwise until the motor linkage disengages. Ensure the plate/motor is supported during the removal.
6. Remove the centrifuge motor from the plate with a 4 mm hex wrench and adjustable wrench (to secure nut).

Note: If reinstalling a motor, when threading the motor/ plate to the linkage, thread clockwise until all the way tight, and then back off slightly to align screws with plate poles (there is one alignment for the poles).

Section 5.15 Valve Removal

Tools Required:

- 18 mm deep well nut driver

1. Remove the front panel (See Section 5.2).
2. Remove the cassette motor plate per Steps 1 – 3 in Section 5.14.
3. Identify the valve to be removed. Unlatch the connector for the valve to be removed.
4. Remove the valve using the 18 mm deep well nut driver.

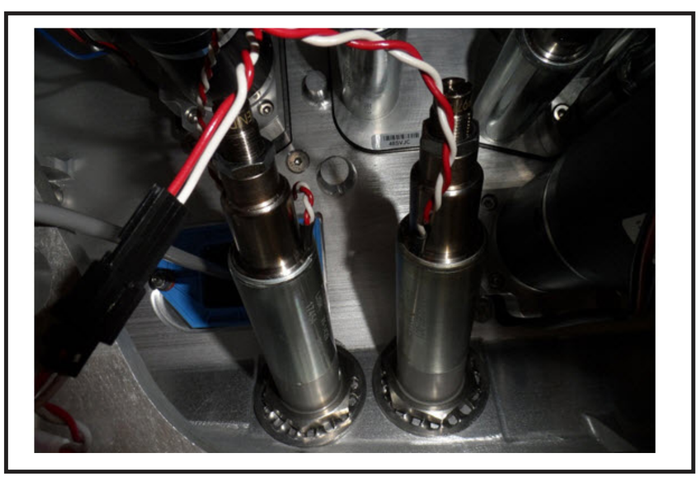
Note: When reinstalling, tighten valve to 150 in-lbs.

Section 5.16 Clamp Removal

Tools Required:

- Clamp removal tool
- PHILLIPS screwdriver

1. Remove the front panel (See Section 5.2).
2. Identify the clamp to be removed. Unlatch the connector to the clamp.



3. Remove the nut holding the latch in place with the clamp removal tool. The tool fits over the wire/clamp and around the retaining nut. Use the PHILLIPS screwdriver in the opening at the end of the tool to turn and unscrew the nut.

4. Pull the clamp up and out of the AmiCORE Apheresis System.

Note: When reinstalling, tighten nut to 266 in-lbs.

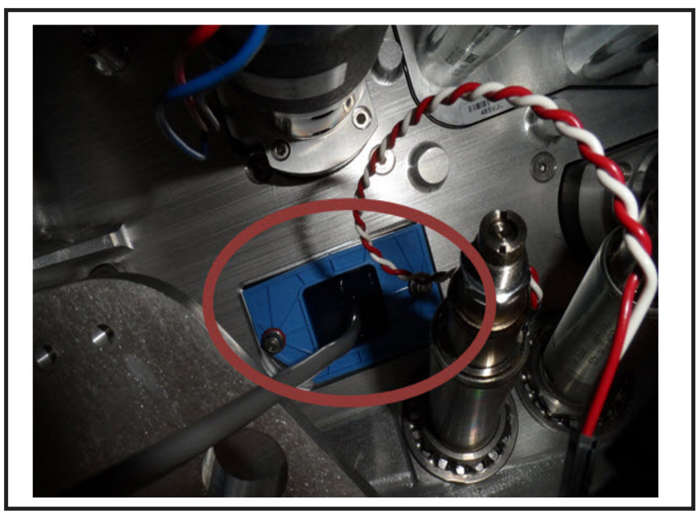
Section 5.17 Air Detector Removal

Tools Required:

- 7 mm nut driver with long and short (ratchet) drives

1. Remove the front panel (See Section 5.2).

2. Locate the air detector. Disconnect the air detector connector.



3. Using the 7 mm nut driver, remove the nuts from the air detector.

4. Remove the air detector from the AmiCORE Apheresis System.

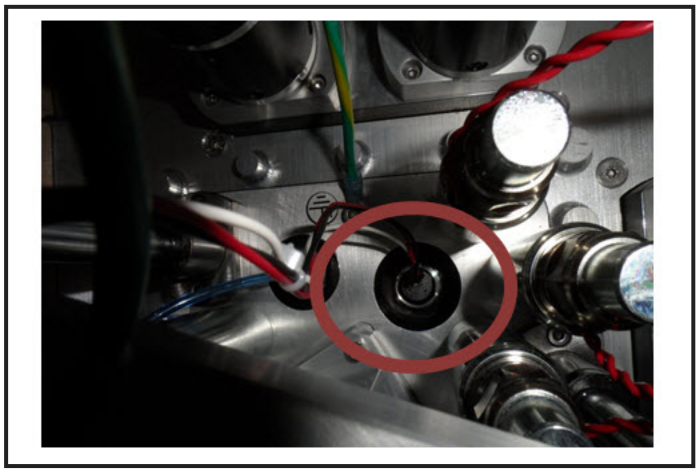
Note: When reinstalling, apply a small bead of LOCTITE 594 or silicon RTV to the outside ridge of the air detector.

Section 5.18 Centrifuge Pressure Sensor Assembly Removal

Tools Required:

- Pressure sensor nut driver tool

1. Remove the front panel (See Section 5.2).
2. Remove the pressure sensor connector from the Main PCBA (either J13 or J14).



3. Remove the pressure sensor with the pressure sensor nut driver tool.

Note: When reinstalling, tighten pressure sensor to 150 in-lbs.

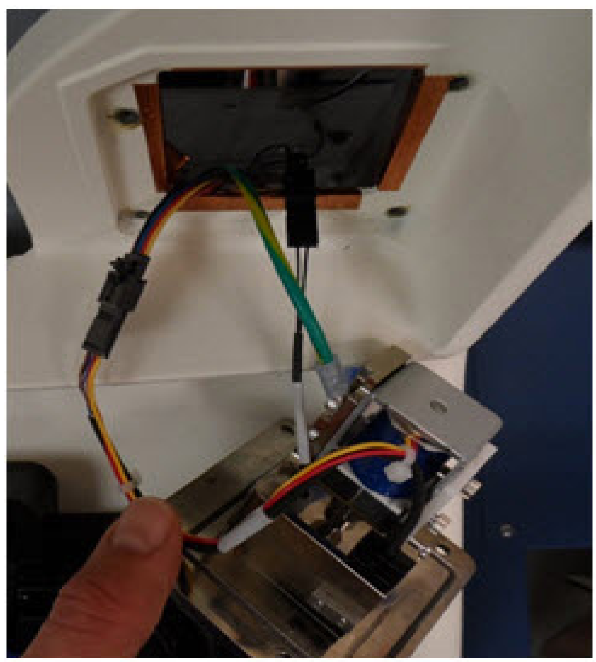
Section 5.19 Door Lock Assembly Removal

Tools Required:

- 2.5 mm hex wrench
- 4.5 mm nut driver

1. Open the centrifuge door.
2. Using the 2.5 mm hex wrench, remove the four screws from the door lock assembly located in the lower right corner of the centrifuge compartment.

3. Slowly pull the door lock assembly out. Disconnect the two connectors. Remove the ground wire from the door lock assembly with the 4.5 mm nut driver. (Suggestion: Place a small piece of tape around the connectors and ground wire to keep them in place for reinstallation of lock assembly.)



4. Remove the door lock assembly from the AmiCORE Apheresis System.

Note: When reinstalling, apply LOCTITE 242 to screw threads. Tighten screws to 9 in-lbs.

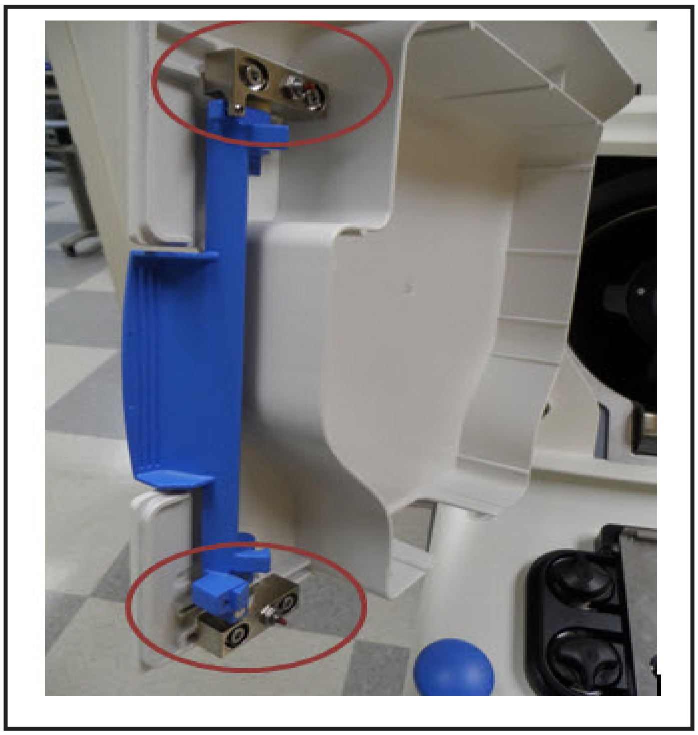
Section 5.20 Door Latch Assembly Removal

Tools Required:

- 4mm hex wrench

1. Open the centrifuge door.

2. Using the 4 mm hex wrench remove the four screws from the door latch assembly located along the left edge of the centrifuge door.



3. Slowly pull the door latch assembly out.

Note: If reinstalling a door latch assembly, ensure spring is clipped into guide rail, loosely tighten screws, close door. Gently pull assembly to the left to engage hooks around lock posts. Open and tighten screws.

4. Test door in locked mode. If door can pull open when locked, adjust set screw (2.5 mm) to improve hook/post connection.

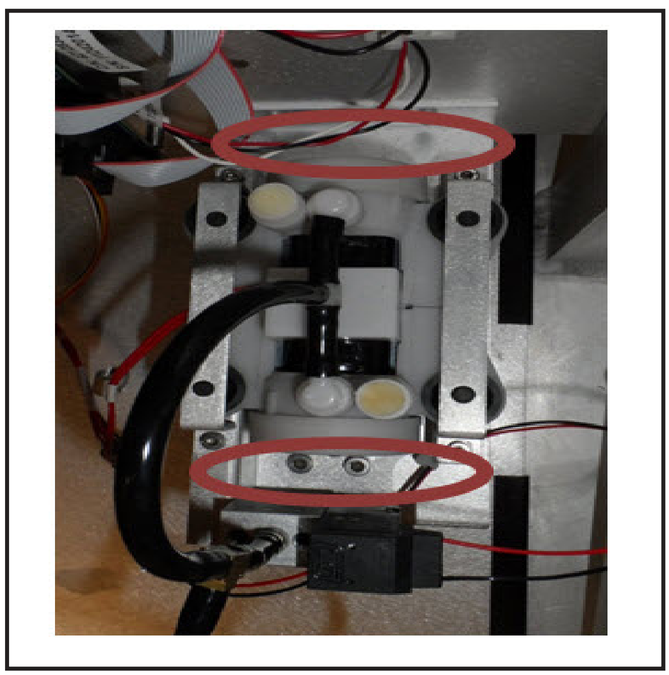
Note: When reinstalling, apply LOCTITE 242 to screw threads. Tighten screws to 44 in-lbs.

Section 5.21 Compressor Assembly Removal

Tools Required:

- 5 mm hex wrench
- Needle nose pliers

1. Remove the rear top panel (See Section 5.3).
2. Remove the rear bottom panel (See Section 5.4).
3. Unplug the compressor connector from the main controller connector J7.
4. Remove the tubing from both sides of the compressor T (the tubing going to the main controller and the tubing going to the compressor port).
5. Using the 5 mm hex wrench, remove the four screws which secure the compressor assembly to the mounting plate.



6. Remove the compressor assembly.

Section 5.22 Leak Detector Removal

Tools Required:

- Standard screwdriver
- Scissors

1. Remove the rear top panel (See Section 5.3).
2. Open the fan frame assembly forward and down.
3. Remove the leak detector connector from the top of the leak detector board. Using the scissors, cut the connector off of the leak detector strip. Remove rubber plug from strip opening and set aside.
4. Carefully peel the leak detector strip off of the centrifuge compartment wall. Pull the remaining portion of the strip through the opening between the centrifuge compartment and the leak detector board.
5. Clean the surface of the centrifuge compartment with an approved fluid per the AmiCORE Operator's Manual.

Section 5.23 Power Supply Distribution Board Removal

1. Remove the rear top panel (See Section 5.3).
2. Remove the rear bottom panel (See Section 5.4).
3. Unplug the connectors along the left edge of the power supply distribution board (main, interface, SBC).
4. Unplug the connectors along the top edge of the power supply distribution board (centrifuge, driver).
5. Unplug the connectors and battery wires from the right edge of the power supply distribution board (battery, display, power).
6. Unplug the two connectors along the lower edge of the power supply distribution board (power supply, power supply).

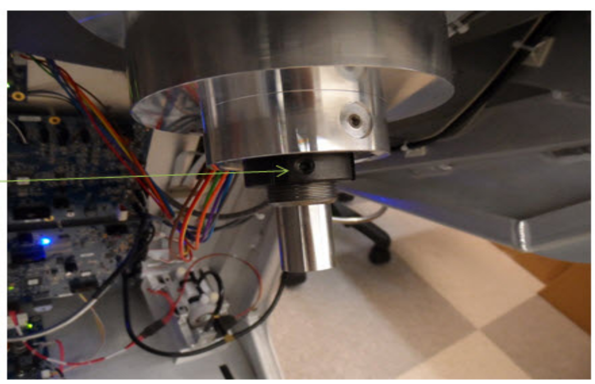
7. Remove the power supply distribution board.

Section 5.24 Centrifuge Assembly Removal

Tools Required:

- Rotor spanner tool
- Ratchet drive
- 3 mm hex wrench

1. Remove the rear top panel (See Section 5.3).
2. Remove the rear bottom panel (See Section 5.4).
3. Loosen the three set screws on the locknut using a 3 mm hex wrench.



4. Using the spanner tool and ratchet drive, loosen the locknut by turning it CCW.
5. Remove the locknut from the shaft.
6. Disconnect the centrifuge connector from the centrifuge board.
7. Remove the rotor from the front of the device.

Note: When reinstalling, apply LOCTITE 242 to screw threads. Tighten set screws to 26 in-lbs. Tighten locknut to 90 in-lbs.

Section 5.25 Centrifuge Motor Removal

Tools Required:

- 6 mm hex wrench
- 4 mm hex wrench

1. Remove centrifuge rotor assembly (See Section 5.24).
2. Remove the six screws holding the motor to the rotor using a 6 mm hex wrench.

3. Disconnect ground wire with the 4 mm hex wrench.
4. Carefully remove the motor from the rotor assembly.

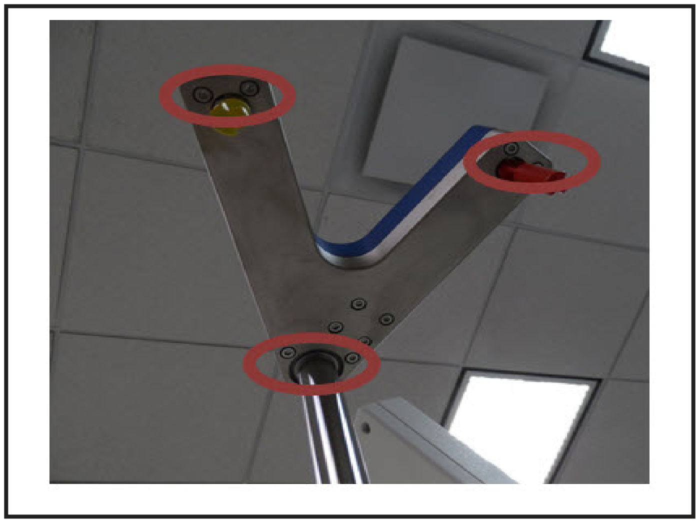
Note: When reinstalling, apply LOCTITE 242 to screw threads. Tighten set screws to 70 in-lbs.

Section 5.26 Scale Assembly Removal

Tools Required:

- 3 mm hex wrench

1. Remove the scale assembly cover using the 3 mm hex wrench. The screws are accessible from the bottom of the scale assembly. Remove the two screws closest to each scale and to the scale assembly pole.



2. Disconnect the scale connection from the scale board.

3. Remove the two mounting screws for the scale. The mounting screws are the two closest to the cable.

4. Carefully remove the scale.

Note: When reinstalling, apply LOCTITE 242 to screw threads. Tighten screws to 26 in-lbs.

Section 5.27 Display Assembly Removal

Tools Required:

- 3 mm hex wrench
- 2.5 mm hex wrench
- 7 mm nut driver

1. Remove the nine screws located in the rear of the display assembly with the 3 mm hex wrench.

2. Disconnect the connector for the daily power button.

3. Remove the display back cover.

4. Disconnect the two cables on the rear of the display.

5. Remove the two harness clamps with the 7 mm nut driver.

6. Carefully remove the four screws holding the display assembly to the hinge latch (two screws per hinge). Secure the display while removing the screws.

7. Remove the display assembly.

Section 5.28 Display Board Removal

Tools Required:

- 3 mm right angle hex wrench.

1. Remove the rear top panel (See Section 5.3).

2. Pull open the fan frame assembly back and down.

3. Remove display cable (J77).

4. Carefully remove the four screws using the 3 mm right angle hex wrench. Secure the board during screw removal.

5. Carefully lower the board and disconnect the three cables along the top edge of the board (J84, J86, J88).

6. Remove the display board.

Section 5.29 Power Supply Removal

Tools Required:

- 4 mm hex wrench
- PHILLIPS screwdriver

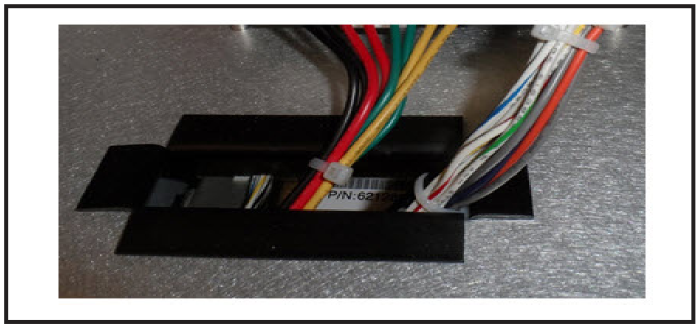
1. Remove the rear top panel (See Section 5.3).

2. Remove the rear bottom panel (See Section 5.4).

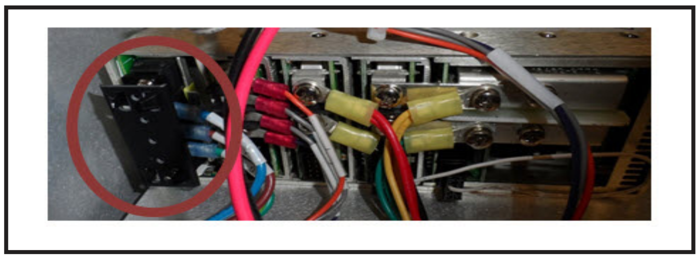
3. Remove all connectors from the power supply distribution board.

4. Loosen the six screws on the power supply compartment top cover. The screws do not have to be completely removed.

5. Slowly remove the power supply compartment. The power supply cables should be fed through the top cover opening during cover removal.



6. Remove the press fit black clip on the left edge of the rear of the power supply. Remove the three wires (from top to bottom: black, brown, green).



7. Remove the six screws on the power supply mounting plate.

8. Remove the power supply mounting plate.

9. Remove the power supply from the mounting plate.

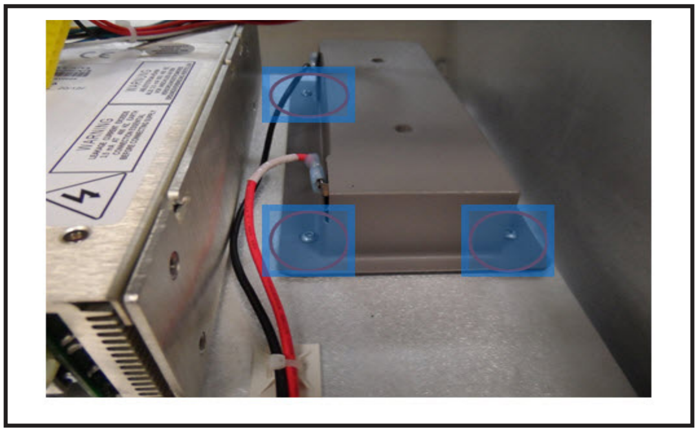
Section 5.30 Battery Removal

Tools Required:

- 2.5 mm hex wrench

1. Follow power supply removal Steps 1 – 5 (See Section 5.29).

2. Remove the four screws from the battery cover.



3. Remove the battery.

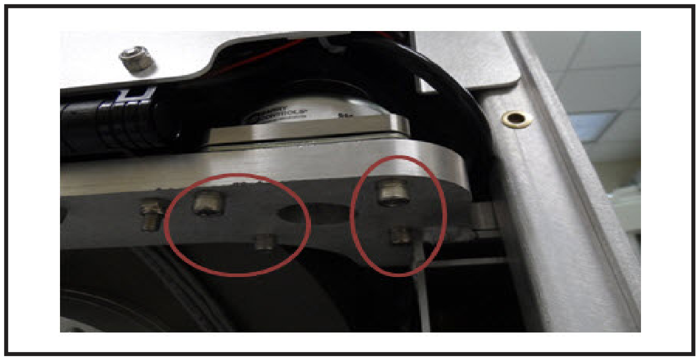
Section 5.31 Isolator Removal

Tools Required:

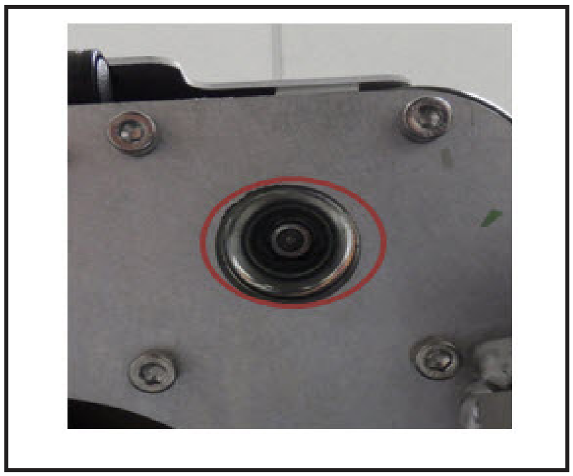
- 4 mm hex wrench
- 5 mm hex wrench
- Adjustable wrench

1. Remove the rear top panel (See Section 5.3).
2. Remove the rear bottom panel (See Section 5.4).
3. Remove the four outside screws below the isolator to be removed using the 4 mm hex wrench.

Note: When reinstalling, tighten screws to 70 in-lbs.



4. Remove the middle screw from the isolator using the 5 mm hex wrench. Secure the top nut with the adjustable wrench during removal.



5. Carefully slide the isolator out.

Section 5.32 Gimbal Removal

1. Open the centrifuge door.

2. Remove the gimbal by gently squeezing on the blue contact points and lifting the gimbal up, back, and out.

Section 5.33 Pressure Sensor O-Ring Removal

Tools Required:

- PHILLIPS screwdriver

1. Remove the three screws from the cassette plate.

2. Remove the cassette plate.

3. Carefully remove the O-ring from the pressure sensor housing.

Note: When reinstalling cassette plate, tighten screws to 26 in-lbs.

Section 5.34 Donor Pressure Sensor Cover Removal

Tools Required:

- PHILLIPS screwdriver

1. Remove the three screws from the cassette plate.

2. Remove the cassette plate.

3. Carefully peel the pressure sensor cover from the pressure sensor housing.

Note: When reinstalling cassette plate, tighten screws to 26 in-lbs.

Section 5.35 Fuse Removal

Tools Required:

- Standard screwdriver

1. Identify the fuse access ports, there are two, they are just above the power cord entry module.

2. Carefully unscrew the fuse access ports with the standard screwdriver.

3. Remove the fuse access ports.

4. Remove the fuses.

Note: If replacing fuses, ensure replacement fuses are supplied by Fresenius Kabi spare parts supply per parts listing. Replace fuses with appropriate fuses for the mains voltage.

Section 5.36 Air Filter Removal

Tools Required:

- PHILLIPS screwdriver

1. Remove the air inlet cover from the front of the instrument with the PHILIPS screwdriver.

2. Remove the air filter from the air inlet cover.

Chapter 6 - Calibration and Adjustment

Introduction

This section describes the procedures to be utilized for calibration or adjustment of specific parts. It contains the following procedures:

- Touch Screen
- Weight Scale Verification and Calibration
- Centrifuge Pressure Sensor Calibration
- Interface Detector Assembly Alignment Calibration
- Interface Detector Signal

Section 6.1 Touch Screen Calibration

1. Begin this procedure with the AmiCORE Apheresis instrument powered OFF.
2. Power the AmiCORE apheresis instrument ON.
3. During the power-up systems test, when the rotating arrow is on the AmiCORE Apheresis System display, touch anywhere on the display quickly five times in a row. This will initiate the Touch Screen calibration procedure.

   Note: If the start-up process reaches the STOP Button Check Screen, it has gone too far, begin this procedure over at Step 1.

4. Follow the on-screen instructions. A small red circle will appear. Touch the center of the circle with your finger.
5. The small red circle will appear at a different location on the screen. Touch the center of the circle with your finger.
6. The small red circle will appear at a different location on the screen. Touch the center of the circle with your finger.

7. Follow the on screen instructions and acknowledge the calibration.

Section 6.2 Weight Scale Verification and Calibration

Tools Required:

- 1 kg weight
- 2.5 kg weight

1. Power the AmiCORE apheresis instrument to ON.
2. From the home screen, touch the *i-button*.
3. Touch the *instrument settings tab* (3rd tab over).
4. Touch the *scale calibration button*.

Verification:

5. Follow the on-screen instructions. Initially the scale performs a 0 g verification check and then a verification check with the 1 kg weight. Place the 1 kg weight on the scale, and touch the *check button*.
6. Remove the 1 kg weight from the scale, and touch the *check button*.
7. If Steps 5 and 6 pass, then the scale is properly calibrated and this procedure is complete. If either step fails, a “!” will appear. If this occurs, the weight on the scale should be verified and the “!” should be acknowledged. If the check fails a second time, scale calibration is initiated.

Calibration:

8. Remove all weight from the scale. Touch the *check button*.
9. Place the 2.5 kg weight on the scale. Touch the *check button*.
10. Remove the 2.5 kg weight from the scale when prompted.
11. Calibration is now complete, repeat Steps 5 – 7 to verify calibration.

Section 6.3 Centrifuge Pressure Sensor Calibration

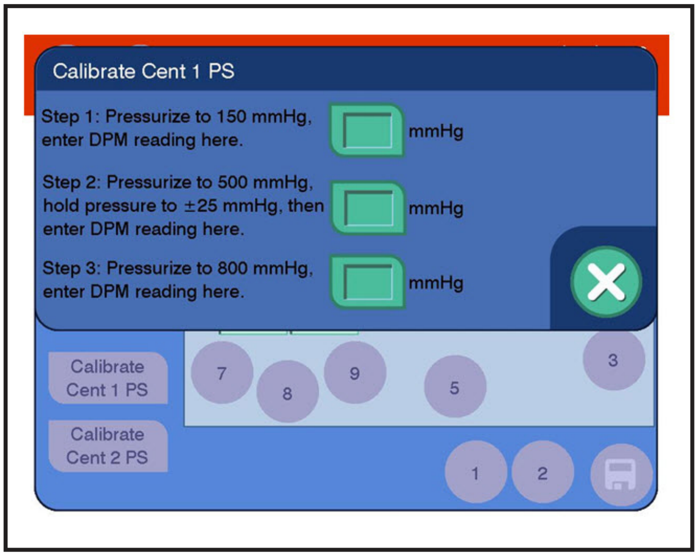
Tools Required:

- Digital pressure meter
- AmiCORE Apheresis System disposable cassette
- Syringe
- Tubing with Y-connector
- Hemostats (2)

1. Power ON the AmiCORE Apheresis System.
2. Enable Service Mode (See Section 3.4).
3. Touch the *pumps tab*.
4. Ensure the *cassette button* reads Disable Motor. If the button reads Enable Motor, touch the button.
5. Place the cassette on the top panel.
6. Set all pumps to -30 RPM.
7. Touch the *down arrow* for the cassette.
8. Stop all pumps. Reverse all pumps at 30 RPM, if necessary, to fully load pump tubing within pump raceway. Reverse again at -30 RPM if necessary.
9. Connect the digital pressure meter to the Y-tubing. Connect the syringe to one of the Y-tubing tubes.
10. Touch the *clamps/PS tab*.
11. Touch the *open clamps button*.

Centrifuge Pressure Sensor #1

12. Attach the Y-tube to the sixth tube over from the top left corner of the cassette.
13. Touch the *calibrate cent 1 PS button* for the pressure sensor to be calibrated.



14. Follow the on-screen instructions. Select the *Step 1 pressure button*. Apply a pressure of approximately 150 mmHg on the digital pressure meter with the syringe. Clamp the line between the syringe and the Y. While holding the pressure enter the DPM reading in the Step 1 pressure keypad, and touch the *check button* to save the value.

15. Select the *Step 2 pressure button*. Apply a pressure of approximately 500 mmHg on the digital pressure meter with the syringe. Clamp the line between the syringe and the Y. While holding the pressure, enter the DPM reading in the Step 2 pressure keypad and touch the *check button* to save the value.

16. Select the *Step 3 pressure button*. Apply a pressure of approximately 800 mmHg on the digital pressure meter with the syringe. Clamp the line between the syringe and the Y. While holding the pressure, enter the DPM reading in the Step 3 pressure keypad and touch the *check button* to save the value.

17. Touch the *X button* on the Calibrate PS screen to lock the calibration in place.

18. Verify pressure sensor calibration by completing the instructions in Section 3.9 for the centrifuge pressure sensor #1.

Centrifuge Pressure Sensor #2

19. Attach the Y-tube to the ninth tube from the top left of the cassette.

20. Touch the *calibrate cent 2 PS button* for the pressure sensor to be calibrated.

21. Follow the on-screen instructions. Select the *Step 1 pressure button*. Apply a pressure of approximately 150 mmHg on the digital pressure meter with the syringe. Clamp the line between the syringe and the Y. While holding the pressure, enter the DPM reading in the Step 1 pressure keypad and touch the *check button* to save the value.

22. Select the *Step 2 pressure button*. Apply a pressure of approximately 500 mmHg on the digital pressure meter with the syringe. Clamp the line between the syringe and the Y. While holding the pressure, enter the DPM reading in the Step 2 pressure keypad and touch the *check button* to save the value.

23. Select the *Step 3 pressure button*. Apply a pressure of approximately 800 mmHg on the digital pressure meter with the syringe. Clamp the line between the syringe and the Y. While holding the pressure, enter the DPM reading in the Step 3 pressure keypad and touch the *check button* to save the value.

24. Touch the *X button* on the Calibrate PS Screen to lock the calibration in place.

25. Verify pressure sensor calibration by completing the instructions in Section 3.9 for the centrifuge pressure sensor #2.

26. Remove the Y-tubing from the cassette.

27. Set all pumps to -30 RPM.

28. Select the *up arrow* for the cassette.

29. Stop all pumps.

30. Remove the cassette.

Section 6.4 Interface Detector Assembly Alignment Calibration

Tools Required:

- Interface Detector Alignment Tool
- Fiber Optic Continuity Tester
- Fiber Optic Cleaning Tool

- 5/32 Right angle hex wrench

1. Power off the AmiCORE Apheresis System.

2. Open the centrifuge compartment door.

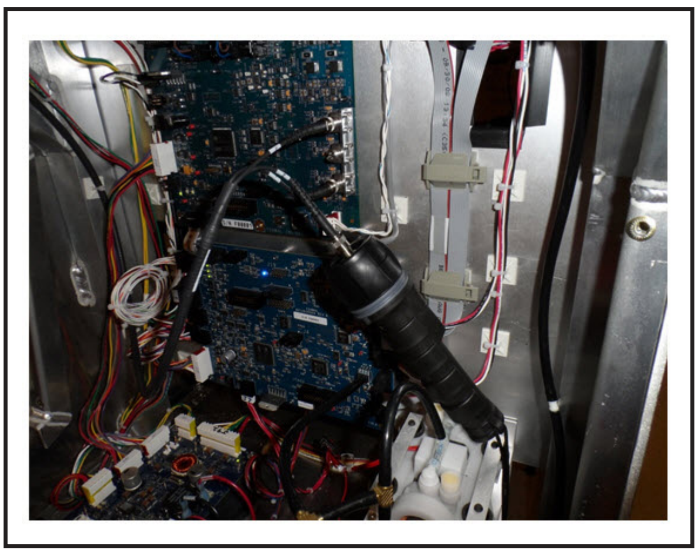
3. Remove the rear top panel by completing the instructions in Section 5.3.

4. Remove the rear bottom panel by completing the instructions in Section 5.4.

5. Using the standard screw driver, open the fan frame assembly. It will swing forward and down on a hinge.

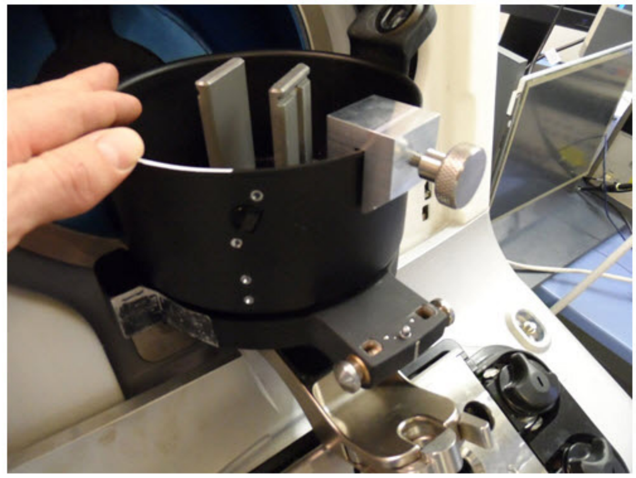
6. Disconnect the fiber optic cable from PD2 of the interface controller board. Do not bend any of the fiber optic cables as bending may break internal fibers.

7. Connect the fiber optic cable to the fiber optic continuity tester. The connection is keyed and threaded. Insert the cable into the key notch and then turn clockwise to secure.



8. Remove the spool from the centrifuge compartment by touching the *spool release latch buttons*.

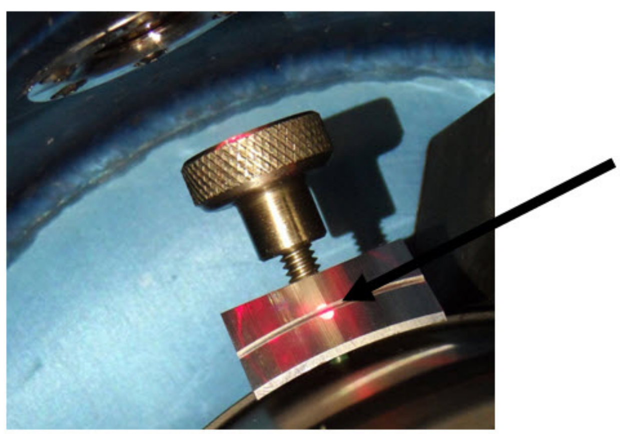
9. Install the interface detector alignment tool on the spool holder. Place the tool as close to the ramp window as possible.



10. Turn the fiber optic continuity tester power ON.

11. Close the centrifuge and manually rotate the spool holder so that the light coming from the interface detector is on the interface detector alignment tool.

12. A properly aligned interface detector assembly will project the light onto the line on the interface detector alignment tool. If the light being projected is centered on the line, then the procedure is complete.



13. If the light being projected is either above or below the line on the interface detector alignment tool, loosen (but do NOT remove) the four screws holding the interface detector assembly with the 5/32 hex wrench. The assembly will now move up and down. Carefully move the assembly while watching the light. Once the light is aligned with

the center line on the interface detector alignment tool, tighten the screws of the interface detector assembly.

14. Disconnect the fiber optic continuity tester and clean the fiber optic cable tip with the fiber optic cleaning tool. Reconnect the cable to PD2. The cable presses into place, and then is threaded to tighten.

Section 6.5 Interface Signal Calibration

Tools Required:

- Fluid filled umbilicus with centrifuge pack

1. Power ON the AmiCORE Apheresis System.
2. Enter Service Mode by completing the instructions in Section 3.4.
3. Touch the *centrifuge tab*.
4. Touch the *door lock button* so that it displays *unlocked*.
5. Open the centrifuge compartment door and install the centrifuge pack and umbilicus in the centrifuge.
6. Close the centrifuge compartment door and touch the *door lock button* so that it displays locked.
7. Touch the *cmd centrifuge button* and enter a commanded centrifuge speed of 1640 RPM.
8. Verify that the centrifuge spins up to speed.
9. Touch the *calibrate interface button*.
10. The AmiCORE Apheresis System will automatically perform the interface calibration procedure. The procedure will automatically update the *brightness and threshold output fields*. Calibration is successful if there are no alerts during the process.

Chapter 7 - Parts List

Section 7.1 Spare Parts Assemblies

Caution: Use only replacement components, cables, and accessories authorized by the manufacturer.

The following is a listing of spare parts assemblies for the AmiCORE Apheresis System. Refer to the AmiCORE Spare Parts List, available via the Global Technical Management internal Sharepoint website, for part numbers and additional information.

Part Name	System
Display Communication PCBA	Electrical
Leak Temperature PCBA	Electrical
Interface Detector Assembly	Electrical
Centrifuge PCBA	Electrical
Interface PCBA	Electrical
Main PCBA	Electrical
Weight Scale PCBA	Electrical
Power Distribution and Battery Charger PCBA	Electrical
Driver PCBA	Electrical
Power Supply Assembly	Electrical
Backup Battery	Electrical
Compressor Assembly	Electrical
Pressure Cuff (regular)	Hardware
Pressure Cuff (large)	Hardware
WS Hook (grey)	Hardware
WS Hook (red)	Hardware
WS Hook (white)	Hardware
WS Hook (yellow)	Hardware
Weight Scale Assembly	Electrical
Leak Detector Strip	Electrical
Fan Assembly	Electrical
Speaker	Electrical
Daily Power Button	Electrical
Emergency STOP Button	Electrical

Part Name	System
Display Assembly	Electrical
Pump Assembly Ratio 1 (Cent, PC, Blood)	Electrical
Pump Assembly Ratio 2 (Recirc, PRP, AC)	Electrical
Donor Clamp (purple)	Electrical
Donor Clamp (blue)	Electrical
Air Detector Assembly Kit	Electrical
Donor Pressure Sensor	Electrical
Centrifuge Pressure Sensor	Electrical
Pump Head Assembly	Mechanical
Valve Assembly	Electrical
Cassette Top Plate Assembly	Mechanical
Cassette Motor Assembly	Electrical
Pressure Sensor PCBA	Electrical
Bar Code Reader	Electrical
Tubing Guide Clip Kit	Hardware
Front Caster	Hardware
Rear Wheel	Hardware
Air Filter	Hardware
Rotor Assembly	Electrical
Centrifuge Motor	Electrical
O Omega Latch Arm Kit	Mechanical
Door Lock Assembly	Electrical
Door Handle Assembly	Mechanical
Door Assembly	Mechanical
Isolator	Mechanical
ID Optical Cable Assembly	Cable
Centrifuge Bowl	Mechanical
Spool	Mechanical
Interface Alignment Tool	Specialty Tool
Screws for Cassette Plate	Hardware
Spool Reflector	Hardware
Wi-Fi Antenna	Hardware
Power Cord US	Electrical
Power Cord EU	Electrical
Fuses 250v 2A (n=2)	Electrical
Cassette Release Button	Electrical
Front Panel Assembly	Hardware
Rear Bumper Assembly	Hardware
Wheel Cover	Hardware
Gimble Assembly	Mechanical

Part Name	System
O Omega Latch Assembly	Mechanical
Weight Scale Pole	Hardware
Stop Bumper O Omega	Hardware
Ball Plunger Detent	Hardware
Bearing Housing Tower Spindle	Mechanical
O Omega Alignment Tool	Specialty Tool
Cover Lower Rear	Hardware
Cover Upper Rear	Hardware

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Chapter 8 - Troubleshooting

Section 8.1 Introduction

The purpose of this chapter is to aid in the assessment and resolution of AmiCORE Apheresis System issues. The troubleshooting information is provided as a reference for trained service personnel. The information is to be implemented in conjunction with service training and systematic troubleshooting techniques when identifying and resolving AmiCORE Apheresis System issues.

The following information is discussed:

- Alerts Listing
- Troubleshooting Tools
- Subsystem Troubleshooting
- Data Downloading – Escalations
- Resolution Verification

Section 8.2 Alerts Listing

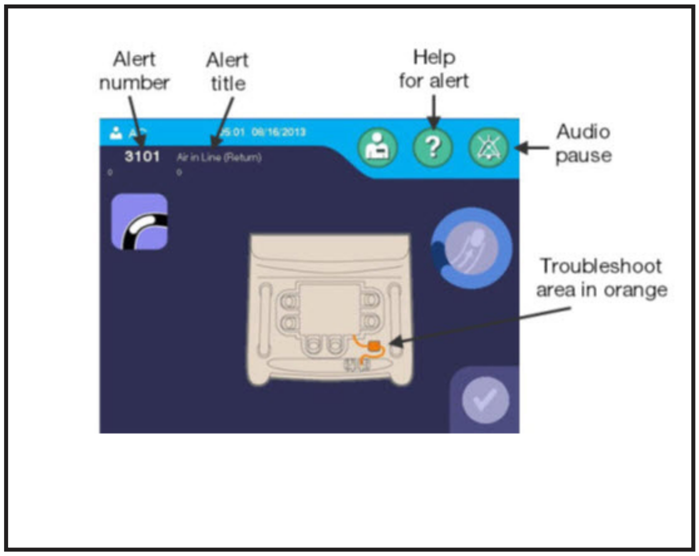
The AmiCORE Apheresis System contains system alerts. The following table identifies each alert contained within the AmiCORE Apheresis System software. The alerts are displayed on the display in the format of: alert number and alert name. The table within this section includes the alert number, the alert name, if the alert is recoverable (R) or non-recoverable (NR), as well as alert trigger conditions and notes (where applicable).

One of the first steps in troubleshooting AmiCORE Apheresis System issues is gathering the details and facts of an issue. In many cases this will be identifying the alert or alerts which the customer was experiencing. Once the alert information is obtained, the table within this section can provide greater details for the alert.

Acquiring the following additional details will assist the process of troubleshooting alerts:

- When, during the procedure, did the alert occur?
- Have there been multiple, confirmed occurrences of the alert on the device?
- For multiple occurrences: is occurrence consistent or intermittent?

Analysis of the alert and the additional information gathered can help identify possible causes of the alert. These possible causes include: operator, disposable kit, hardware failure, other.



Code	Description	Resolution	Data 1	Data 2
1002	Cannot connect to network The system could not connect to the DXT Relay server.	Check that the Ethernet cable is plugged in or the wireless network is functional.	0	0
1029	Fluid Detected During the initial instrument self-tests, the system detected fluid at the air detector.	If kit from the previous procedure is installed, remove the kit. Press the STOP *button* and shut off instrument. If previous kit was not on instrument when alert was triggered, air detector may be malfunctioning. Call a qualified service representative.	0	0
1030	Stop Button Error During the initial instrument self-tests, the STOP button test failed.	Power off instrument and call a qualified service representative.	1 -PRESSED 2 -NOT_PRESSED	1 – Main 2 – Safety
1040	Calibration Failure The system detected a front scale error.	Touch the *calibrate scale button* to recalibrate the front scale. Ensure weight is hung properly on the front weight scale. If alert persists, press the STOP *button*. Power off the instrument and call a qualified service representative.	0	Gain*1000
1041	Scale Disturbed The system was unable to collect a stable scale reading.	Check the scales and touch the *check button* to continue. Call a qualified service representative if the issue persists.	0	0
1042	Weight on Scale The system detected an incorrect weight on the front scale.	Remove any weight from front scale. If no weight is on scale, touch the *calibrate scale button* and perform scale calibration. Touch the *check button* to continue. If issue reoccurs, a hardware error is likely. Call a qualified service representative.	Ave Gross Weight1 (Front Scale)	0

Code	Description	Resolution	Data 1	Data 2
1043	Calibration Failure The system detected no weight change during front scale calibration. Touch the *calibrate scale button* to restart calibration.	Touch the *calibrate scale button* to redo scale calibration. Ensure that during scale calibration, the correct weights are hung on the front scale.	0	0
1044	Calibration Failure The system detected a back scale error.	Touch the *calibrate scale button* to recalibrate the back scale. Ensure weight is hung properly on the back weight scale. If alert persists, press the STOP *button*. Power off the instrument and call a qualified service representative.	0	Gain*1000
1045	Weight on Scale The system detected an incorrect weight on the back scale.	Remove any weight from back scale. If no weight is on scale, touch the *calibrate scale button* and perform scale calibration. Touch the *check button* to continue. If issue reoccurs, a hardware error is likely. Call a qualified service representative.	Ave Gross Weight2 (Back Scale)	0
1046	Calibration Failure The system detected no weight change during back scale calibration.	Touch the *calibrate scale button* to redo scale calibration. Ensure that during scale calibration, the correct weights are hung on the back scale.	0	0
1101	Power Failure Instrument experienced power failure upon power on. Check that the instrument is plugged in.	Press the STOP *button*. Power OFF the instrument at both power switches. If issue persists, call a qualified service representative.	0	0

Code	Description	Resolution	Data 1	Data 2
1102	Battery Charge Failed The system detected an error with the battery charge circuit.	Keep instrument plugged in and power switches ON to allow battery to charge. Touch the *check button* to continue. Repeat to allow battery to charge. Call a qualified service representative if the issue persists.	0	0
2002	Kit Check Error The blood pump has pumped fluid without seeing a pressure change, indicating an occlusion problem with the draw line clamp, return line clamp, or saline clamp (inside the cassette).	Touch the *check button* to continue. Ensure proper alignment of the cassette and tubing within clamps before reloading the cassette.	0	0
2003	Kit Check Error The saline clamp is not functioning or fluid has been introduced; probable cassette load error.	Press the STOP *button* and end the procedure. Start over with new kit. Do not spike fluids until instructed to.	0	Donor 1 PS pressure delta
2004	Kit Check Error Blood pump is unable to generate sufficient pressure to perform saline prime; due to air leak or faulty saline solution connection.	Check all connections and tubes between saline container and blood pump for poor connections; fix connections and touch the *check button*. If no connection issues exist, inspect for air leak. If air leak present, press the STOP *button* and end the procedure. Start over with new kit.	0	Blood Pump Volume, PC Pump Volume

Code	Description	Resolution	Data 1	Data 2
2005	Kit Check Error The blood pump has pumped fluid without seeing a change in pressure; the manual donor clamp is not properly occluding, the AC pump is not loaded correctly, or there is an air leak.	Check that donor line manual clamp is in fully closed position. Touch the *check button* to continue. If issue persists, check that tube is properly oriented around AC pump and touch the *check button* If issue persists, air leak is likely the cause. Press the STOP *button* and end the procedure. Start over with new kit.	0	0
2007	Kit Check Error The system is unable to build pressure within the kit.	Touch the *check button* to continue. Ensure proper alignment of the cassette and clamps before reloading the cassette.	0	0
2008	Kit Check Error The system is unable to build pressure within the kit.	Touch the *check button* to continue. Ensure proper alignment of the cassette and clamps before reloading the cassette.	0 1	0 Donor 2 pressure delta
2010	Leak Detected The system has detected a leak in the kit.	Press the STOP *button* and end the procedure. Unload kit and start again with new kit.	0	0
2011	Kit Check Error Draw line clamp or return line clamp is not loaded properly.	Touch the *open clamps button* to correctly install the draw and return lines. Touch the *check button* to continue. Ensure proper alignment of the cassette and clamps before reloading the cassette.	0	0
2020	Leak Detected The system detected a leak in the kit during kit install check.	Press the STOP *button* and end the procedure. Unload kit and start over with a new kit.	1	Donor 1 pressure delta

Code	Description	Resolution	Data 1	Data 2
2024	Install Kit Timeout The kit has been loaded on instrument for too long without confirmation of venipuncture; dry kit cannot be loaded on instrument for more than eight hours and wet kit cannot be loaded on instrument for more than four hours.	A dry kit cannot be loaded on instrument for more than eight hours and a wet kit cannot be loaded on instrument for more than four hours. Press the STOP *button* and end the procedure. Unload kit and start over with a new kit.	0	0
2025	Kit Check Error The system is unable to build pressure within the kit.	Touch the *check button* and re-install kit. If issue persists, hardware error likely. Call a qualified service representative.	1	Donor 2 pressure delta
2026	Leak Detected The system detected a leak.	Press the STOP *button* and end the procedure. Unload kit and start over with a new kit.	0	0
2027	Plasma Container off Back Scale The system detected insufficient weight on the back scale.	Place the plasma container on the back scale. Touch the *check button* to continue.	0	Ave gross weight 2 (back scale)
2028	Extra Weight on Back Scale The system detected excessive weight on the back scale.	The plasma container should be the only weight on the back scale. Remove any other weights and touch the *check button* to continue.	0	Back scale weight delta
2029	Containers Not on Front Scale The system detected insufficient weight on the front scale.	The in-process and RBC containers should be the only weights on the front scale. Remove any other weights and touch the *check button* to continue.	0	Weight on front scale

Code	Description	Resolution	Data 1	Data 2
2030	Extra Weight on Front Scale The system detected excessive weight on the front scale.	Place only the in-process, RBC containers on the front scale. Touch the *check button* to continue.	0	Weight on front scale
2101	Pressure Out of Range During solutions prime, pressure in the disposable set was outside of the allowable range.	Inspect kit and lines for kinks. Once fixed, touch the *check button*. If issue persists, hardware error likely. Call a qualified service representative.	1 2 3 4	Donor 1 Inst pressure Donor 2 Inst pressure Centrifuge 1 Inst pressure Cent 2 Inst pressure
2102	Saline Prime Error The saline container is not connected properly or the air detector tubing is misloaded.	Ensure saline line is not kinked. Ensure the saline container has been spiked properly and is hanging on the solution pole on the right side. If there is fluid in the air detector tubing, the tubing should be removed and reinserted in order to be fully loaded in the air detector. Touch the *check button* to continue	0	Blood pump volume
2103	AC Prime Error The AC container is not connected properly.	Spike the AC container, fix connection, or check for kinks. Touch the *check button* to continue.	0	Blood pump volume
2104	Saline Container on Scale The saline container is on one of the scales.	Remove the saline container from the weight scale and hang from solution pole on right side. Touch the *check button* to continue.	0	Blood pump volume
2105	Saline Prime Error The draw and return lines are interchanged in clamps.	Touch the *open clamp button*. Reload tubing, switching so that inlet line is in its clamp and return line is in its clamp. Touch the *check button* to close clamps and continue.	0	0

Code	Description	Resolution	Data 1	Data 2
2106	AC Container Not on Scale AC container is not on the scale.	Place the AC container on the front scale. Touch the *check button*.	0	Kit Check – AvgNetWeight1 (Front Scale) Fluid Prime - Weight delta during AC line prime
2107	AC Prime Error Flow issue detected with AC container; the AC container is not loaded properly, there is a kink in the AC line, or the AC container is empty.	Ensure AC container is loaded correctly. Inspect AC line for kinks. If AC level is low, touch the *add container button* to add a new AC container. Touch the *check button* to continue.	AC Pump Correction Factor *100	1 – Correction factor greater than allowable maximum 2 – Correction factor smaller than allowable minimum
2108	Saline Prime Error Flow issue detected with saline container; the saline container is not installed properly, there is a kink in the saline line, or the saline container is empty.	Ensure saline container is loaded correctly. Inspect saline line for kinks. If saline level is low, touch the *add container button* to add a new saline container. Touch the *check button* to continue.	Blood Pump Correction Factor *100	1 – Correction factor greater than allowable maximum 2 – Correction factor smaller than allowable minimum
2109	Air in Line (Prime) Air detector has detected air where there should be liquid; a leak is present, the saline container is empty, or the appropriate line has become misaligned with the air detector.	Check that line is properly aligned with air detector. Check for saline in saline container and touch the *add container button* to replace if low. Inspect for leaks.	0 – the system will pump a small volume of saline and try the check again. 1 – the system will try the check again.	0
2110	Air Detector Failed Air detector has failed.	Hardware failure, call a qualified service representative.	0	0

Code	Description	Resolution	Data 1	Data 2
2111	Interface Detection Issue During Prime, the system found the interface detector (optical sensor) was outside of the allowable range.	Touch the *check button* to retry calibration. Check interface detector window and ramp for debris. Clean by soaking one edge of a lens cloth with isopropyl alcohol and wipe the interface detector window and ramp gently, in a circular motion, with the alcohol soaked edge of the lens cloth. Continue to wipe gently until the parts are completely clean. Call a qualified service representative if the issue persists.	1 – Centrifuge spinning when commanded to unlock 2 – Interface calibration took too long 3 – Centrifuge not commanded to max speed 4 – Pulse width 1 out of range 5 – Pulse width 2 out of range 6 – Pulse width 3 out of range 7 – Bad samples taken, centrifuge not at full speed 8 – LED brightness out of range	1 – 0 2 – 0 3 – 0 4 – Interface pulse width 1 5 – Interface pulse width 2 6 – Interface pulse width 3 7 – Speed error samples 8 – LED brightness signal
2113	Leak Detected During prime, there was a leak in the kit or clamps are not working properly.	Press the STOP *button* and end the procedure. Unload kit and start over with a new kit.	1 - Unable to build pressure at clamp 6/7 2 - Unable to build pressure at clamp 6/7 3 - Unable to build pressure at clamp 9 4 - Unable to build pressure at clamp 8 5 - Unable to build pressure at clamp 10	Centrifuge 2 Inst gauge pressure
2114	Incorrect Container on Front Scale An incorrect AC container weight was detected on the front scale.	Ensure that the volume of the AC container loaded is equivalent to the AC container volume displayed on the Hang AC screen/ overlay. Touch the *check button* to continue.	AvgNetWeight1 (Front Scale)	0
2115	Saline Prime Error There is an occlusion in the container lines or centrifuge lines.	Check for kinks or damaged tubing within the kit. Ensure that there is nothing disturbing the scale. Touch the *check button* to continue.	Centrifuge Pump Correction Factor *100	1 – Correction factor greater than allowable maximum 2 – Correction factor smaller than allowable minimum

Code	Description	Resolution	Data 1	Data 2
2117	Platelet Clamp Closed The clamp on the platelet container line is closed and should be open.	Open the platelet line clamp if it is closed. Ensure the platelet container line is not kinked. Touch the *check button* to continue.	0	Pressure at platelet clamp
2118	Plasma Container on Front Scale The system detected the plasma container on the wrong scale.	Place the plasma container on the back scale. Touch the *check button* to continue.	Weight delta on front scale during centrifuge pump calibration	Pump volume delta during centrifuge pump calibration
2119	IP Container on Back Scale The system detected the in-process container on the wrong scale.	Place the in-process container on the front scale. Touch the *check button* to continue.	BloodPumpCorrectionFactor*100	Weight on front scale before blood pump calibration
2120	RBC Container on Back Scale The system detected the RBC container on the wrong scale.	Place the RBC container on the front scale. Touch the *check button* to continue.	Weight delta on back scale	Weight delta on front scale
2121	Plasma Container Not on Scale The system detected the plasma container on the wrong scale.	Place the plasma container on the back scale. Touch the *check button* to continue.	0	Weight delta on back scale
2122	IP Container Not on Scale The system detected the in-process container is not on the scale.	Place the in-process container on the front scale. Touch the *check button* to continue.	- Blood Pump Volume - Weight on Front scale	- Weight on front scale before blood pump calibration - Minimum weight required on front scale

Code	Description	Resolution	Data 1	Data 2
2123	RBC Container Not on Scale The system detected the RBC container is not on the scale	Place the RBC container on the front scale. Touch the *check button* to continue.	Weight delta on back scale	Weight delta on the front scale
2124	Clamp Check Error The system was unable to relieve pressure in the kit.	Inspect kit and lines for kinks. Once fixed, touch the *check button*. If issue persists, hardware error likely. Call a qualified service representative.	6 – Clamp 6 not venting 7 – Clamp 7 not venting 8 – Clamp 8	Centrifuge 2 Inst gauge pressure
2125	Close Donor Clamp The system cannot draw a negative pressure against the manual donor clamp. Verify this clamp is closed.	Check that the donor roller clamp is closed. Inspect for leaks. Touch the *check button* to continue. If issue persists, a leak may be present in the kit.	0	Donor 1 Ave pressure
2301	Pressure Out of Range During inter-procedure self-tests, one or more of the pressure sensors was not at atmospheric pressure.	Ensure nothing is resting on pressure sensors on the top panel prior to selecting a procedure. Check the sensors for debris and clean by using compressed air, if necessary. If blue hydrophobic filters on donor pressure sensors are wet, replace them. Touch the *check button* to continue. Call a qualified service representative if the issue persists.	1 – Donor 1 out of range 2 – Donor 2 out of range 3 – Cent 1 out of range 4 – Cent 2 out of range	Donor 1 Ave pressure Donor 2 Ave pressure Cent 1 Ave pressure Cent 2 Ave pressure
2306	Weight on Scale During inter-procedure self-tests, a weight was detected on the front scale.	If a container is on the front scale, remove and touch the *check button*. If no containers were on the scale, touch the *calibrate scale button* and follow instructions to calibrate the scale.	1	AveGrossWeight1 (front scale)

Code	Description	Resolution	Data 1	Data 2
2307	Weight on Scale During inter-procedure self-tests, a weight was detected on the back scale.	If a container is on the back scale, remove and touch the *check button*. If no containers were on the scale, touch the *calibrate scale button* and follow instructions to calibrate the scale.	2	Ave Gross Weight2 (back scale)
2310	Cuff Pressure Out of Range During inter-procedure self-tests, the cuff pressure was outside of atmospheric range.	Check that cuff is properly deflated. Detach and vent any residual pressure. Touch the *check button* to continue.	0	Cuff Ave pressure
2312	Leak Detector Signal Error During inter-procedure self-tests, the leak detector signal was outside threshold.	Press the STOP *button* and touch *shut down*. Power off instrument and call a qualified service representative.	Leak Sensor 1 Reading Leak Sensor 2 Reading	1 2
2313	Temperature Out of Range During inter-procedure self-tests, the temperature was outside of atmospheric range.	Ensure that the instrument is within specified operating temperatures. Move instrument from possible heat sources. Touch the *check button*. Ensure instrument air filters are clean and unobstructed. Call a qualified service representative if the issue persists.	Temperature Sensor 1 Reading Temperature Sensor 2 Reading	1 2

Code	Description	Resolution	Data 1	Data 2
3003	Interface Detector Blocked The interface detector signal was lost.	Check the interface detector window and ramp for debris or smudges. Clean by soaking one edge of a lens cloth with isopropyl alcohol and wipe the interface detector window and ramp gently in a circular motion, with the alcohol soaked edge of the lens cloth. Check the centrifuge pack and plasma container for lipemic plasma. If lipemic plasma is present, and the alert is retriggered, stop and end the procedure.	0	0
3004	Scale and Pumps Out of Sync There was a change in the weight on the scale that was inconsistent with the pump direction.	Inspect for kinks in lines. Inspect for pump misalignment. Fix appropriate issue and touch the *check button*.	0	0 – Blood Prime 2 – Draw 4 – Reinfusion 5 – Air Purge 6 – Reprime
3005	Low Saline Container Volume The saline container volume is too low.	Touch the *add container button* to add a new saline container. Touch the *check button* to continue.	0	0
3006	Low AC Container Volume The AC container volume is too low.	Touch the *add container button* to add new AC container. Remove the old AC container before adding a new AC container. Touch the *check button* to continue.	0	0
3007	Draw Volume Limit Exceeded The system reached the maximum draw volume limit.	Press the STOP *button* and end procedure. Disconnect donor. Unload kit.	0	0

Code	Description	Resolution	Data 1	Data 2
3008	Pressure Out of Range Pressure in the kit was out of range.	Check the kit for kinks in the tubing lines. Touch the *check button* to continue.	3 4	Cent 1 Inst gauge pressure Cent 2 Inst gauge pressure
3009	Pressure Out of Range Pressure in the kit was out of range.	Check the kit for kinks in tubing lines. Touch the *check button* to continue.	1 2 3 4	Donor 1 Inst gauge pressure Donor 2 Inst gauge pressure Cent 1 Inst gauge pressure Cent 2 Inst gauge pressure
3010	Pressure Out of Range Pressure in the kit was out of range.	Check the kit for kinks in tubing lines. Touch the *check button* to continue.	0	0
3011	Container Volume Limit Exceeded The system reached the maximum container volume limit.	Press the STOP *button* and end procedure. Disconnect donor. Unload kit.	0	0
3012	Draw Volume Limit Exceeded The system reached the maximum draw volume limit.	Press the STOP *button* and end procedure. Disconnect donor. Unload kit.	0	0
3101	Air in Line (Return) Air was detected in the donor line during fluid return.	The system will automatically perform an air purge. Check that the return line is properly inserted into the air detector. Following the air purge, check the return line for air and touch the *check button* if it is enabled. If air remains in line or *check button* is still disabled, touch the *air purge button* to perform a second air purge. Repeat if necessary. Once air has been cleared, touch the *check button* to continue.	0	0

Code	Description	Resolution	Data 1	Data 2
3102	Repeat Cent. Line Blockage A second consecutive centrifuge line occlusion was detected without purging the fistula of stagnant blood. Consider performing saline reinfusion.	Check the kit for kinks in the tubing lines. Touch the *check button* to continue. When in collection, touch the *pause button* and return saline to the donor in order to flush the fistula.	Number of centrifuges inlet line occlusions in procedure	Volume drawn since the last centrifuge inlet line occlusion
3103	Centrifuge Line Blockage The system detected a subsequent attempt to clear a centrifuge line occlusion.	Check the kit for kinks in the tubing lines. Touch the *check button* to continue.	Number of centrifuges inlet line occlusions in procedure	Volume drawn since the last centrifuge inlet line occlusion
3104	Blood Prime Error Volume of whole blood drawn during blood prime is more than max volume that can be drawn.	Press the STOP *button* and end the procedure. Disconnect donor. Unload kit. Donor must be deferred; do not start over with new kit.	1 – > Draw Cycle Limit 2 – > Total Volume Out Limit 3 – Weight on Scale >Total	Blood pump volume
3105	Air in Line (Draw) The system detected air in donor inlet line repeatedly during the blood prime.	Check venipuncture and line in air detector. Make sure manual clamp on donor line is open. Check for kinks on donor line. Check that the tubing is properly aligned in the air detector. Touch the *check button* to continue.	0	0

Code	Description	Resolution	Data 1	Data 2
3508	No Blood Flow Vein pressure fell below the minimum pressure during a draw phase.	Check the inlet line for kinks or occlusions. Check venous access and position of the donor's arm. Instruct donor to squeeze when the cuff inflates. Check that the cuff on the inlet line is inflated to the appropriate pressure. Touch the *decrease value button* to decrease the inlet rate. Check that the donor roller clamp is open. Touch the *check button* to continue. If issue persists, consider lowering the inlet rate further.	0	0
3601	High Pressure Return Vein pressure exceeded the maximum allowed pressure during a return phase.	Check the donor's return venipuncture site for signs of infiltration. Check the return line for kinks or occlusions. Touch the *decrease value button* to decrease the return rate. Touch the *check button* to continue. If the issue persists, consider lowering the return rate further.	0	0
4002	Plasma Collection Error Insufficient plasma was collected.	Touch the *check button* to continue.	0	Plasma for end product
4003	Weight on Scale The system has detected an increase in the weight on the scale indicating that the operator has hung the centrifuge pack on one of the scales.	Remove centrifuge pack from scale and touch the *check button*.	0	0

Code	Description	Resolution	Data 1	Data 2
4004	Weight on Scale The system has detected an increase in the weight on the scale indicating that the operator has hung the platelet container on one of the scales.	Remove platelet container from scale. Touch the *check button* to continue.	0	0
4005	Product Transfer Error The system detected a high pressure in the centrifuge pack.	Unload the centrifuge pack from the centrifuge and hang on the right solution pole. If the pack is already unloaded, check the kit for kinks in the tubing lines.	0	0
4007	Product Transfer Error PC pump and scale are out of sync during product transfer.	Inspect tubes for kinks. Check for an empty plasma container. Rectify and touch the *check button*.	PC pump volume	Plasma on scale
8001	Software Error A SW communication error occurred.	Press the STOP *button* and end procedure. Disconnect donor. Unload kit. Shut down the instrument and call a qualified service representative.	0	0
9001	Software Error The system was unable to capture the procedure record.	Touch the *check button* to continue. Call a qualified service representative if the issue persists.	0	0
10001	Scale Disturbed The scales sensed a large, sudden change in weight.	Check that nothing is disturbing or moving the containers on the scale. Steady the containers. Touch the *check button* to continue. If problem persists, evaluate surroundings for sources of disturbance (construction, etc.).	WS Variance 1 (front scale)	WS Variance 2 (back scale)

Code	Description	Resolution	Data 1	Data 2
10002	Power Failure The system detected a loss of AC power.	Plug the instrument's power cord into the outlet and touch the *check button* to continue. If instrument power is lost for five minutes, the instrument will shut down.	0	0
10003	Incorrect Weight on Scale The weight on the scale is incorrect. Ensure that the correct containers are on the scale.	Check that the correct containers are on the scale. Check the scale for any disturbances.	1 2	Weight delta on front scale Weight delta on back scale
10004	Air Detector Failed The system detected a fault within the air detector circuit.	Press the STOP *button* and end the procedure. Disconnect donor. Call a qualified service representative.	1 – Air detector reading air incorrectly during test 2 – Air detector reading fluid	0
10005	Pump Speed Error The measured pump speed did not match the commanded pump speed.	Check pumps for obstructions. Touch the *check button* to continue. Call a qualified service representative if the issue persists.	1 – AC Pump 2 – Blood Pump 3 – Centrifuge Pump	1 2 3 4
10006	Pump Speed Error The pump head is failing to turn when a non-zero rate has been commanded.	Check pumps for obstructions. Ensure pumps are all the same height. If a pump is higher than the others, reinstall the pump as described in Section 6.4. Touch the *check button* to continue. Call a qualified service representative if the issue persists.	0	1 – AC Pump 2 – Blood Pump 3 – Centrifuge Pump 4 – PC Pump 5 - PRP Pump 6 - Recirc Pump

Code	Description	Resolution	Data 1	Data 2
10007	Pump Direction Error The pump head is not turning in the commanded direction.	Press the STOP *button* and end the procedure. Disconnect donor.	0	1 – AC Pump 2 – Blood Pump 3 – Centrifuge Pump 4 – PC Pump 5 – PRP Pump 6 – Recirc Pump
10008	Pump Speed Error The AC pump is turning faster than the blood pump.	Press the STOP *button* and end the procedure. Disconnect donor.	0	0
10010	Pump Speed Error The AC pump is turning during a return cycle.	Press the STOP *button* and end the procedure. Disconnect donor.	0	0
10011	Cassette Error Cassette had been unloaded following a power failure.	Press the STOP *button* and end the procedure. Disconnect donor.	0	0
10012	Pump Volume Error Pump head volume is not consistent with pump motor volume.	Check pumps for obstructions. Touch the *check button* to continue. Call a qualified service representative if the issue persists.	1 – AC Pump 2 – Blood Pump 3 – Centrifuge Pump 4 – PC Pump 5 – PRP Pump	0
10013	Auto Recovery Failed Instrument pump speed auto-recovery retries exceeded.	Touch the *check button* to continue. If the issue persists, call a qualified service representative.	0	0

Code	Description	Resolution	Data 1	Data 2
10014	Clamp Error The clamps did not respond in time.	Touch the *check button* to continue. If issue persists, call a qualified service representative.	Clamp 1 Clamp 2 Clamp 3 Clamp 4 Clamp 5 Clamp 6 Clamp 7 Clamp 8 Clamp 9 Clamp 10 Clamp 11	0
10015	Pump Volume Error The system detected inconsistent pump volume.	Press the STOP *button* and end procedure. Disconnect donor. Unload kit.	1 – AC Pump 2 – Blood Pump 3 – Centrifuge Pump 4 – PC Pump 5 – PRP	0
10110	Centrifuge Speed Error The centrifuge speed was slower than expected.	Check for obstructions. Touch the *check button* to continue. Call a qualified service representative if the issue persists.	0	0
10111	Centrifuge Speed Error The centrifuge speed was faster than expected.	Press the STOP *button* and end the procedure. Disconnect donor. Call a qualified service representative.	1 – Centrifuge spinning faster than first safety tolerance, but below second safety tolerance. 2 – Centrifuge	Measured centrifuge speed

Code	Description	Resolution	Data 1	Data 2
10115	Air Purge Malfunction Either the saline manual clamp is closed, the saline container is empty, or there is a problem with the saline line.	Inspect for kinks. Ensure manual saline clamp is open. Check that the tubing is properly aligned in the air detector. Check saline container level. Touch the *add container button* to add a replacement container. Touch the *check button* to continue.	1 – Donor 1 PS indicates saline bag empty or kinked 2 – Repeated air seen during air purge 3 – Not enough volume pumped	0
10116	Blood Prime Error The system detected insufficient flow from the donor.	Check the inlet line for kinks or occlusions. Check venous access and position of the donor's arm. Instruct donor to squeeze when the cuff inflates. Check that the cuff on the inlet line is inflated to the appropriate pressure. Check that the donor roller clamp is open.	0	0
10117	Centrifuge Door Not Closed System has detected that the centrifuge door is not closed.	Close centrifuge door and touch the *check button* to continue. If door is already closed, open it and close it again. Touch the *check button* to continue	0	0
10118	Centrifuge Door Error Centrifuge door did not successfully lock before timeout was reached.	Open and re-shut centrifuge door. Touch the *check button* to continue. Call a qualified service representative if the issue persists.	0 – Door took too long to lock 1 - Door remained locked for too long	0
10120	Pump Speed Error A pump speed issue was detected.	Touch the *check button* to continue. Call a qualified service representative if the issue persists.	1 – AC Pump 2 – Blood Pump 3 – Centrifuge Pump 4 – PC Pump	0
10121	Clamp Over Current The system detected a pump error.	Touch the *check button* to continue. Call a qualified service representative if the issue persists.	When combined with data 2, forms a bit-packed value where a 1 is in the X-bit position where ClampXOverCurrent = FAULT.	See Data 1

Code	Description	Resolution	Data 1	Data 2
10122	Pump over current The system detected a pump error.	Touch the *check button* to continue. Call a qualified service representative if the issue persists.	Bit-packed value with 1 in the bit positions that corresponds to the fault as follows: bit 1 – AC pump bit 2 – Blood pump bit 3 – PRP pump bit 4 – PC pump bit 5 – Recirc pump bit 6 – Cent pump	0
10123	Pump Fault The system detected a pump error.	Touch the *check button* to continue. Call a qualified service representative if the issue persists.	Bit-packed value with 1 in the bit positions that corresponds to the fault as follows: bit 1 – AC pump bit 2 – Blood pump bit 3 – PRP pump bit 4 – PC pump bit 5 – Recirc pump bit 6 – Cent pump	0
10124	Centrifuge Fault The system detected a centrifuge error.	Touch the *check button* to continue. Call a qualified service representative if the issue persists.	Measured rpm of the centrifuge motor at the time of the fault.	0
10125	Centrifuge Speed Error The system detected a centrifuge speed error.	When the centrifuge has stopped, check the door lock for any obstructions. Touch the *check button* to continue. Call a qualified service representative if the issue persists.	1	Commanded centrifuge rpm

Code	Description	Resolution	Data 1	Data 2
10126	Pressure Out of Range Pressure in the disposable set was outside the expected range.	Check the kit for kinks in the tubing lines. Check that the platelet line clamp is open. Touch the *check button* to continue.	1 – Donor 1 2 – Donor 2 3 – Cent 1 4 – Cent 2	Donor 1 inst gauge pressure Donor 2 inst gauge pressure Cent 1 inst gauge pressure Cent 2 inst gauge
11002	Return Pathway Error The system detected that fluid was returned through the inlet line.	Press the STOP *button* and end the procedure. Disconnect donor.	0	0
11003	Draw Pathway Error The system detected that fluid was drawn through the return line.	Press the STOP *button* and end the procedure. Disconnect donor.	0	0
11004	Blood Prime Error Insufficient volume was drawn into kit during blood prime. Possible presence of air.	Press the STOP *button* and end the procedure. Disconnect donor.	1 – Insufficient blood was drawn during reprime 2 – Insufficient blood was drawn before returning to the donor	0
11005	Power Failure The system has detected a power failure.	Press the STOP *button* and end the procedure. Disconnect donor. Power off instrument and call a qualified service representative.	Power reading	1 +5 digital monitor 2 +5 Analog monitor 3 +15 Monitor 4 -15 monitor 5 +24 Monitor Main 6 +24 Monitor Driver
11006	Hardware Error Commanded safe state was not achieved.	Press the STOP *button* and end the procedure. Disconnect donor. Power off the instrument.	0	Bit-packed value with 1 in the bit positions that corresponds to the fault by numerical order of the clamps

Code	Description	Resolution	Data 1	Data 2
11007	AC Ratio Error The system detected an incorrect measured AC ratio above allowed maximum.	Press the STOP *button* and end the procedure. Disconnect donor.	0	0
11008	Temperature Out of Range Exceeded centrifuge temperature limit for one of the centrifuge temperature sensors.	Check that nothing is blocking the air inlet filter or the air outlet paths. Remove any obstructions. Ensure that the instrument is within specified operating temperatures. Move instrument from possible heat sources. Touch the *check button*. If issue persists, press the STOP *button* and end the procedure. Disconnect donor.	Temperature reading	1 – Temperature sensor 1 2 – Temperature sensor 2
11010	Cuff Error Cuff pressure has gone above the maximum pressure limit.	Touch the *check button* to continue. If the issue persists, contact a qualified service representative.	0	Cuff inst gauge pressure
11011	Cassette Error Cassette unloaded while the donor was connected.	Press the STOP *button* and end the procedure. Disconnect donor. Call a qualified service representative.	0	0
11012	AC Ratio Error The system detected a measured AC ratio below the safety minimum	Press the STOP *button* and end the procedure. Disconnect donor.	0	0
11013	Cuff Error The pressure cuff was inflated during a return cycle.	Check the cuff tubing connection. Touch the *check button* to continue. If the issue persists, call a qualified service representative.	0	Cuff inst gauge pressure
11014	AC Ratio Error AC ratio error was detected during draw.	Press the STOP *button* and end the procedure. Disconnect donor.	0	0

Code	Description	Resolution	Data 1	Data 2
11015	Air Purge Malfunction Insufficient air purge volume pumped.	Press the STOP *button* and end the procedure. Disconnect donor.	Air message reading	Safety air message reading
11017	Hardware Error A software error preventing HW safe state was detected.	Disconnect donor. Power off the instrument.	STOP button main reading	STOP button safety reading
11018	Software Error The system detected a software error.	Press the STOP *button* and end the procedure. Disconnect donor. Power OFF the instrument. Call a qualified service representative.	0	0
11020	Air Detector Failed The system detected a fault within the air detector circuit.	Press the STOP *button* and end the procedure. Disconnect donor. Call a qualified service representative.	Safety air message reading	Safety air pulse message reading
11021	Pressure Out of Range The system detected a fault with a pressure sensor.	Press the STOP *button* and end the procedure. Disconnect donor. Call a qualified service representative.	1 – Donor 1 2 – Donor 2 3 – Centrifuge 1 4 – Centrifuge 2	Safety monitor pressure reading for the pressure sensor
11022	Centrifuge Leak Detected The system detected a leak in the centrifuge.	Press the STOP *button* and end the procedure. Disconnect donor.	Leak sensor 1 reading	Leak sensor 2 reading
11023	Leak Detector Signal Error The system detected a fault with the leak detector.	Press the STOP *button* and end the procedure. Disconnect donor. Call a qualified service representative.	Leak sensor 1 reading	Leak sensor 2 reading
11024	Software Error The system detected a software error.	Press the STOP *button* and end the procedure. Disconnect donor.	1 2	0

Code	Description	Resolution	Data 1	Data 2
11025	Return Pathway Error The system detected that fluid was pumped into the saline container.	Press the STOP *button* and end the procedure. Disconnect donor.	0	0
11026	Temperature Out of Range Both temperature sensors exceeded centrifuge temperature limit.	Press the STOP *button* and end the procedure. Disconnect donor.	Temperature sensor 1 reading	Temperature sensor 2 reading
11027	Blood Prime Error Saline clamp open while return line is primed.	Press the STOP *button* and end the procedure. Disconnect donor.	0	0
11028	Hardware Error The inlet or return line clamp is open during an alert while the donor is connected.	Press the STOP *button* and end the procedure. Disconnect donor. Power OFF the instrument.	0	0
11029	Software Error The system detected a software error.	Press the STOP *button* and end the procedure. Disconnect donor. Call a qualified service representative	0	0
11030	Centrifuge Speed Error The system detected an error with the centrifuge door lock or door lock sensor.	Press the STOP *button* and end the procedure. Disconnect donor. Call a qualified service representative.	2	Measured centrifuge speed
11031	Software error The system detected a software error.	Press the STOP *button* and end procedure. Disconnect donor. Shut down the instrument. Call a qualified service representative.	Donor connected flag	Donor connected status flag

Code	Description	Resolution	Data 1	Data 2
11032	Centrifuge speed error The centrifuge spin time reached the maximum limit.	Press the STOP *button* and end procedure. Disconnect donor. Shut down the instrument. Contact a qualified service representative.	Inst measured centrifuge speed	0
11033	Cassette Error The system detected a cassette motor error.	Press the STOP *button* and end procedure. Disconnect donor. Unload kit. Call a qualified service representative if the issue persists.	1 – Cassette disabled but Cassette motor status remains enabled. 2 – Cassette motor status is enabled when the donor is connected.	0
11070	Cuff Error The system detected a cuff inflation error.	Cuff will vent pressure. Touch the *check button* to continue. If alert occurred before venipuncture, perform venipuncture to avoid a second occurrence of this alert.	1	Average cuff pressure
14011	Software Error The system detected a software error.	Press the STOP *button* and end the procedure. Disconnect donor. Power OFF the instrument. Contact a qualified service representative.	0	0
14012	Software Error The system detected a software safety check failure.	Press the STOP *button* and end the procedure. Disconnect donor. Power OFF the instrument. Contact a qualified service representative.	0	0
14013	Software Error The system detected a software safety check failure.	Press the STOP *button* and end the procedure. Disconnect donor. Power OFF the instrument. Contact a qualified service representative.	0	0
15001	Software Error The system detected a software error.	Press the STOP *button* and end the procedure. Disconnect donor. Power OFF the instrument. Contact a qualified service representative.	0	0

Code	Description	Resolution	Data 1	Data 2
15002	Software Error The system detected a software error.	Press the STOP *button* and end the procedure. Disconnect donor. Power OFF the instrument. Contact a qualified service representative.	0	0
15003	Software Error The system detected a software error.	Press the STOP *button* and end the procedure. Disconnect donor. Power OFF the instrument. Contact a qualified service representative.	0	0
15004	Software Error The system detected a software error.	Touch the *check button* to continue. If the issue persists, call a qualified service representative.	0 – SW spin 1 – No conversions being completed 2 – Partial conversions are being completed	For data1 = 1 or 2, count of the number of occurrences of each alert on the given power cycle.
15005	Software Error The system detected a software error.	Touch the *check button* to continue. If the issue persists, call a qualified service representative.	Copy cat was triggered	0

Section 8.3 Troubleshooting Tools – Hardware Issues

When troubleshooting the AmiCORE Apheresis System issues, there are multiple tools or aids which may assist in issue-identification.

Interconnect Diagram

The interconnect diagram located in Chapter 2 is a hardware identification tool to utilize when troubleshooting instrument hardware issues. This interconnect diagram represents the relationship between: physical components (clamps, pumps, centrifuge, etc.), printed circuit boards, and system cables. Once a hardware issue is identified, the interconnect diagram may be utilized to narrow down said issue to a specific subsystem. Focus may then be turned to said subsystem for identifying the specific component(s) responsible for the issue.

Diagnostics Screens

The diagnostic screens contained within the AmiCORE Apheresis System Service Mode are the primary tools used in troubleshooting instrument hardware issues. The diagnostic screens are identified and explained in Section 2.7. These screens are utilized to control and test specific components and subsystems of the instrument. The screens are also utilized in the verification of changes implemented during the process of repairing hardware issues.

Simulated Procedure

The simulated run is a troubleshooting tool which may be utilized to:

- verify and recreate alerts which occurred during normal customer operation.
- verify that changes implemented during the process of hardware repair were effective.

See Chapter 3 for detailed instructions on performing a simulated procedure.

Section 8.4 Subsystem Troubleshooting Hardware Issue

Once an issue is narrowed down to possible hardware issue AND the AmiCORE subsystem affected is identified using the tools identified in Chapter 8 of the Service Manual, the next step is to narrow the issue down to the specific component(s).

Pumps, Clamps, and Valves Subsystems

The AmiCORE Apheresis System design utilizes multiple components of the same part for several subsystems. For example, each of the cassette valves is the same part number. This also applies to the clamps and pumps (there are three each of two different pump motor ratios).

Therefore, when an issue has been narrowed down to the pump, clamp, or valve subsystem, the issue can be further diagnosed by changing connections within the system with known good parts. This process can take some time given the complexity of some subsystems; however, it can help narrow down issues further as well as ensure repair effectiveness.

Example

A hardware issue has been narrowed down to the centrifuge pump subsystem. That subsystem consists of: pump, cable, and driver board. To troubleshoot the issue further, the cable connection going to the pump can be connected to the PC pump. This will now change the subsystem to PC pump ("good pump") and original cable and driver board. Using the test menu to command the centrifuge pump on, the trained service provider can verify if the issue moved to the PC pump (indicating the issue is with either the original cable or the driver board) or did not move to the PC pump (indicating the centrifuge pump was faulty). If the issue moved, then a similar test can be performed by using a different pump cable and returning to original centrifuge pump and driver board. That test should help verify whether the cable or the driver board was the root cause.

This process of elimination, utilizing a single change with a known good component, will help narrow down subsystem issues with pumps, clamps, and valves.

Other Subsystems

For subsystems which do not have multiple components of the same part, such as the air detector or display module, it is good troubleshooting practice to inspect cables and connectors. If cables and connectors look good, try reseating the cables and verifying if the issue is resolved.

Section 8.5 Data Downloading – Issue Escalation

The AmiCORE Apheresis System stores procedure data within its computer memory. Issues requiring additional support resources and/or escalations to individuals off site may require the downloading and emailing (or file transfer using approved channels) of the instrument system data. This section outlines the procedure to retrieve AmiCORE Apheresis System data files.

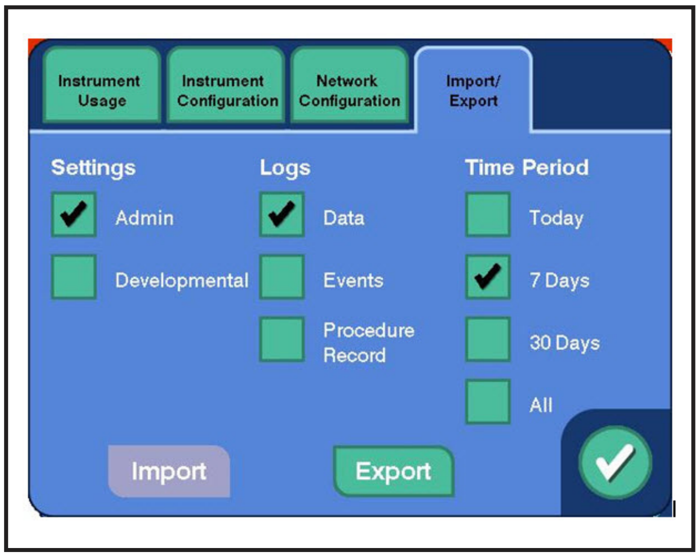
Note: Data recorded while performing tests in service mode may also be downloaded; select the *disk button* prior to downloading.

1. Place an AmiCORE USB drive in one of the AmiCORE USB ports.

2. Place the AmiCORE Apheresis System in Service Mode per Section 3.4.

3. Touch the *i-button*.

4. Touch the *import/export tab*.



5. Select the *developmental button*. A check mark will appear.

6. Touch the *data, events,* and *procedure record buttons*. Check marks will appear next to all three.

7. Select the *all button*. A check mark will appear.

8. Touch the *export button*.

9. The data files will download to the AmiCORE USB drive.

Section 8.6 Resolution Verification

The final step in troubleshooting, planning, and resolution of AmiCORE Apheresis System issues is verification of implemented fix (when applicable). Always follow the appropriate service procedures when performing any repair. The following tests may help in conjunction with, or in addition to, service repair procedure testing.

Service Mode Subsystem Testing

The Service Mode diagnostic screens are useful tools when verifying implemented repairs to the instrument. Place the instrument in service mode. Reference the specific screens and options (see Chapter 2). The Service Mode screens enables the service technician to command specific components.

Example: After a repair is made to the centrifuge pump subsystem, such as replacement of the driver board, the Service Mode may be used to command the pump to specific speeds to verify the repair was effective. In addition, Service Mode may be used to command other components controlled by the replaced driver board to ensure other subsystems were not affected during the repair and that all driver board connections were reconnected correctly.

AmiCORE Field Repair Test Matrix

The AmiCORE Field Repair Test Matrix identifies required tests which must be performed when replacing specific instrument parts.

Simulated Procedure

A simulated procedure is a troubleshooting tool which may be utilized to verify that changes implemented (part replacement) during the process of hardware repair were effective. See Chapter 3 for detailed steps on executing a simulated procedure.

Note: A simulated procedure performed with demo mode enabled does not check the interface detector. Therefore, if the interface detector requires verification, perform Section 3.13 in addition to the simulated procedure.

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